

Nucleophilic substitution reaction

Learning outcomes

At the end of this section you should be able to:

- (R3.4.1) Recognise the nucleophile in chemical reactions.
- (R3.4.2) Deduce equations with descriptions and explanations of the movement of electron pairs in nucleophilic substitution reactions.

When you learn and deduce the chemical equation between reactants and products (see [section R2.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/)), you need to rearrange reactant molecules to form new chemical bonds and produce product molecules. This will only happen when the reactant molecules share or transfer their electron pairs between them. Molecules that have a pair of electrons will be attracted to the positively charged regions on other molecules forming a new covalent bond. Do you think this will happen randomly? As you learn more about the fundamentals of electron pairs specifically in organic chemistry reactions, you will be able to understand the interaction of electrons between the reactant molecules.

Nucleophiles in chemical reactions

[Subtopic S2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) covers the covalent model, which involves a shared pair of electrons between two atoms. The formation of bonds occurs due to the tendencies of attracting electrons to gain stability by sharing their valence electrons. Additionally, a similar process happens in the [coordination bond formation](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/) but the sources of a pair of electrons are shared from the same atom (see [section S2.2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/)).

Look at the molecules in **Figure 1**.

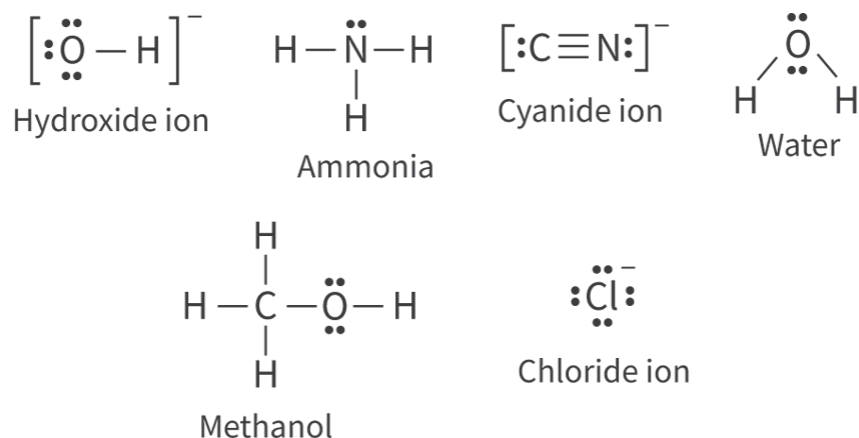


Figure 1. Several anions and neutral molecules.

More information for figure 1

The image is a diagram displaying the structures of various anions and neutral molecules. It includes the following chemical structures:

1. Hydroxide ion - depicted as "[O-H]-" with lone pairs on the oxygen.
2. Ammonia - depicted as "H-N-H" with three hydrogen atoms bonded to nitrogen.
3. Cyanide ion - depicted as "[C≡N]-" indicating a triple bond between carbon and nitrogen.
4. Water - depicted as "H-O-H" with lone pairs on oxygen.
5. Methanol - depicted as "H-C-O-H" with additional hydrogen atoms bonded to carbon.
6. Chloride ion - depicted as "[Cl]-" with lone pairs of electrons shown around the chlorine.

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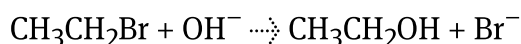
What do these molecules have in common and how do these molecules differ from each other?

You may notice that some of them are neutral (ammonia, water and methanol) and others are anions (hydroxide ion, chloride ion and nitrite ion). However, they all have at least one lone pair of electrons that are

available to form a new covalent bond with other atoms or molecules.

Anions with an extra electron cause them to more readily form a chemical bond with an electron-deficient species. Molecules that are shown with the non-bonding pairs of electrons above are examples of nucleophiles. A nucleophile is defined as an electron-rich species (neutral molecule or negatively charged ion) that has a tendency to donate a pair of non-bonding electrons to an electron-deficient species to form a new covalent bond.

Nucleophiles play an important role in many chemical reactions, including substitution reactions. For example, in the reaction between bromoethane and hydroxide ions.



What happens to a lone pair belonging to the hydroxide ion?

The non-bonding pairs of electrons on the hydroxide ion act as a nucleophile and are donated to the partially positively charged carbon atom of the C–Br bond. The electronegativity difference between carbon and bromine causes the carbon atom to become partially positively charged. As a result, the bromide ion is replaced by the hydroxide ion which produces ethanol and a bromide ion.

This process also occurs similarly to neutral species such as water.



Water as a nucleophile donates a pair of electrons to the partially positively charged carbon atom of the C–Br bond and replaces the bromine atom, forming ethanol and hydrogen bromide.

Common nucleophiles that are used in organic chemistry reactions: neutral molecules with a lone pair of electrons such as alcohols, water (H₂O), ammonia (NH₃) and negatively charged ions that have a lone pair of electrons such as halide ions (Cl[−], Br[−], I[−]), hydroxide ions (OH[−]) and cyanide ions (CN[−]).

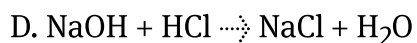
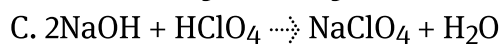
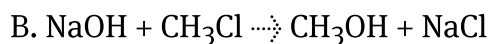
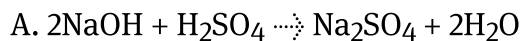
Study skills

The cyanide ion, CN[−], is a nucleophile that contains a triple bond between carbon and nitrogen atoms. This type of reaction is useful, as any halogenoalkanes reacting with cyanide ions will increase the length of the

carbon chain.

Worked example 1

From the reactions shown below,



which one involves NaOH acting as a nucleophile?

In reaction B, the hydroxide donates a lone pair of electrons to the partially positively charged carbon atom of the chloromethane, producing methanol and sodium chloride.

Reaction of A, C and D involves neutralisation where NaOH is acting as a base.



Exercise 1

Click a question to answer



1. Which of the following molecules acts as the nucleophile in the reaction below?



Check answers

Try the activity in **Interactive 1** to check your understanding of the key terms used in this section,

Drag the words into the correct boxes

Nucleophiles often participate in a variety of chemical reactions such as substitution reactions. A nucleophile is an _____ species and is typically negatively charged ion or _____ molecules that have a non-bonding pair of electrons. This available pair of electrons is donated to an _____ species and forms a new _____ bond. For example, a reaction between iodomethane (CH_3I) and a cyanide ion produces acetonitrile (CH_3CN) and an iodide ion. In this reaction the cyanide ion acts as a _____ attacking the _____ atoms in the iodomethane and replacing the iodine atoms.

covalent

electron-deficient

carbon

nucleophile

electron-rich

neutral

✓ Check

Interactive 1. Nucleophilic Substitution Reactions.

The movement of the electron pair in nucleophilic substitution reaction

In a chemical reaction, a nucleophile donates electron pairs to a positively charged region of a molecule (called an electrophile, which will be explained in more detail in [section R3.4.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/)) as shown in

Figure 2. This type of reaction is called a nucleophilic substitution reaction: Nu is nucleophilic, R is electrophile and X is the leaving group.

The movement of electron pairs in the chemical reactions is shown using 'curly arrows'. These are commonly used in organic reaction mechanisms to show the movement of electrons between reactants or between a nucleophile and a substrate, including when the electrons are pulled away to the leaving group that is being replaced.

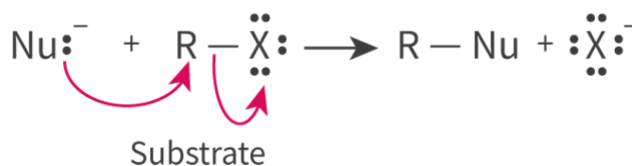


Figure 2. Equation of nucleophilic substitution reaction.

 More information for figure 2

The diagram illustrates a nucleophilic substitution reaction. On the left, the image shows a nucleophile (Nu:) interacting with a substrate (R-X) where R is a carbon chain and X is a leaving group. Curly arrows are used to indicate the movement of electron pairs: one arrow originates from the nucleophile pointing towards the carbon (R), while another curly arrow starts from R-X, showing the electrons moving away from the X, which is labeled as the leaving group. This results in the formation of a new compound on the right, R-Nu, where the nucleophile has replaced the leaving group X, now depicted with a pair of dots and a negative charge. The diagram explains the electron flow during the substitution process using the curly arrows to indicate bond formation and cleavage.

[Generated by AI]

These curly arrows help to illustrate how bonds are broken and formed during a chemical reaction. The destination of a pair of electrons is shown by the head of the curly arrow and the tail shows the source of a pair of electrons. The movement of a pair of electrons is represented by the direction of the arrow. For example, see the reaction of bromoethane and sodium hydroxide in **Figure 3**. Note that the sodium hydroxide dissociates in solution to produce the hydroxide ion, OH⁻, which then goes on to react with the bromoethane.



Figure 3. Nucleophilic substitution reaction of bromoethane with sodium hydroxide.

 More information for figure 3

The image is a diagram illustrating a nucleophilic substitution reaction. It shows hydroxide ion (:OH^-) reacting with bromoethane ($\text{C}_2\text{H}_5\text{-Br}$). The hydroxide ion acts as a nucleophile, indicated by a curly arrow pointing from :OH^- to the carbon atom in $\text{C}_2\text{H}_5\text{-Br}$, which is partially positively charged. This leads to the formation of ethanol ($\text{C}_2\text{H}_5\text{-OH}$) and the bromide ion (Br^-). The bromine atom in bromoethane is shown as the leaving group with a pair of electrons, indicated by another curly arrow pointing from the bond between carbon and bromine toward the bromine atom, which is then stabilized as Br^- .

[Generated by AI]

The hydroxide ion from sodium hydroxide acts as a nucleophile and attacks carbon because it is partially positively charged, and replaces bromine as the leaving group (**Figure 4**). A lone pair of hydroxide ions is donated to the carbon atom forming a new bond between carbon and hydroxide to produce ethanol. The leaving group is considered more stable than the nucleophile as it is less reactive and able to retain its electronic configuration by pulling back its electrons when they are being replaced by the nucleophiles.

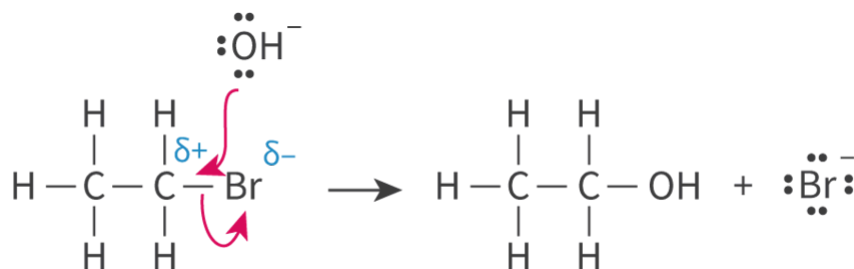


Figure 4. The (hydroxide ion) nucleophile attacks carbon on bromoethane.

 More information for figure 4

The image is a chemical diagram illustrating a nucleophilic substitution reaction. It shows the molecular structures before and after the reaction. On the left side, bromoethane is depicted, with a partially positively charged carbon atom (δ^+) bonded to bromine, which has a partial negative charge (δ^-). A hydroxide ion (:OH^-) is shown approaching the carbon atom, with an arrow indicating the direction of the attack. The arrow starts from the lone pair on the hydroxide ion and points towards the carbon-bromine bond, illustrating the substitution process. After the reaction arrow, on the right side,

ethanol (H-C-C-OH) is displayed as the product along with bromine ion (:Br⁻), indicating the completion of the reaction. The diagram captures the flow of electrons and the change in bonding resulting from the substitution.

[Generated by AI]

In general, a nucleophile is more likely to play a part in a chemical reaction depending on its reactivity. The more reactive the nucleophile, the more easily it donates electrons. This is usually affected by the electronegativity of the nucleophile, the use of solvent for the reaction that takes place and the electrophile that the nucleophile is reacting with.

The role of nucleophiles in chemical reactions typically occurs by following one of several mechanisms that will be explained in more detail in the HL section R3.4.9–10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/>). This depends on the nature of both the donor and acceptor of a pair of electrons including the reaction conditions.

Study skills

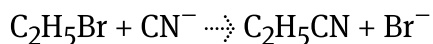
Nucleophiles do not attack positively charged molecules randomly. Their interaction occurs depending on the specific and predictable manner referring to the chemical properties of both molecules that donate and accept the electrons. In this way, you can predict the possible outcomes.

Try **Worked example 1** and the exercise question on the role of nucleophiles in chemical reactions.

Worked example 1

Deduce the chemical equation for the reaction between bromoethane and the cyanide ion. Explain their interaction and the movement of electron pairs in that reaction.

The chemical equation for the reaction between bromoethane and the cyanide ion is as follows:



The movement of a lone pair of electrons on the cyanide ion approaches the partially positive carbon atom of the C–Br bond. The carbon–bromine bond then breaks, causing the bromine atom to depart from the molecule as a leaving group taking both electrons from the carbon–bromine bond, forming a bromine ion. A new bond between the carbon atom and the cyanide ion is formed, producing propanenitrile.



Exercise 2

Click a question to answer



1. Which of the following reactions correctly shows the electrons movement when methanol as a nucleophile reacts with chloromethane?



Check answers

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication — Negotiate ideas and knowledge
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Your task

Pair up with another student and write the products of the chemical equations below. Include the electron pairs of the nucleophile, substrate and the products, as well as the curly arrow to show the movement of the electrons (refer to **Figure 2**).

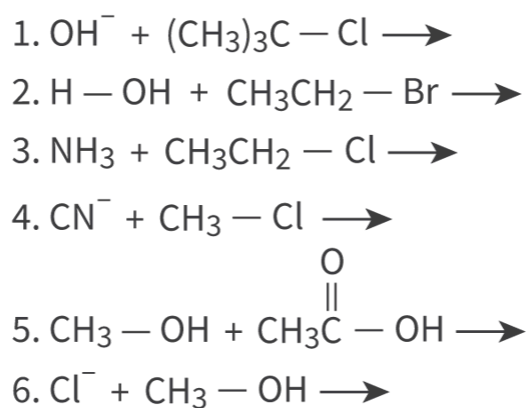


Figure 5. Write the products of these chemical equations.

 More information for figure 5

The image contains six numbered chemical equations featuring various reactions requiring the identification of reaction products, electron pairs, and electron movement through a curly arrow notation.

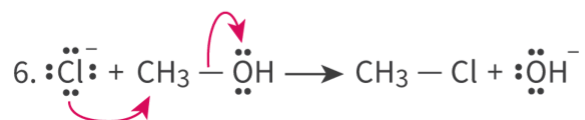
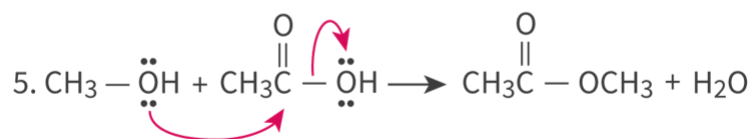
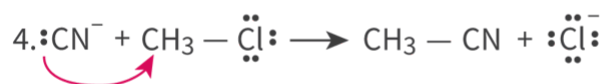
1. Equation 1: $\text{OH}^- + (\text{CH}_3)_3\text{C} - \text{Cl} \rightarrow$
2. Equation 2: $\text{H} - \text{OH} + \text{CH}_3\text{CH}_2 - \text{Br} \rightarrow$
3. Equation 3: $\text{NH}_3 + \text{CH}_3\text{CH}_2 - \text{Cl} \rightarrow$
4. Equation 4: $\text{CN}^- + \text{CH}_3 - \text{Cl} \rightarrow$
5. Equation 5: $\text{CH}_3 - \text{OH} + \text{CH}_3\overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH} \rightarrow$ (with a double-bonded oxygen on top of the second carbon)
6. Equation 6: $\text{Cl}^- + \text{CH}_3 - \text{OH} \rightarrow$

Each reaction includes reactants like hydroxide, water, ammonia, cyanide ion, and chloride ion, paired with various organic substrates like tert-butyl chloride and ethanol, requiring prediction of the products and demonstrating the arrow mechanisms to account for electron shifts.

[Generated by AI]

When you are finished, discuss with your partner what kind of difficulties you encountered to complete this activity? Is there any specific concept you need to highlight?

The mark scheme is available [here](#).



Heterolytic fission

Learning outcomes

At the end of this section you should be able to:

- Explain, with equations, the formation of ions by heterolytic fission.
- Explain the relationship between heterolytic fission and nucleophile.

What happens to the electrons when a covalent bond is broken? A covalent bond is formed between a shared pair of electrons. Thus, when the bond breaks, the electrons are supposedly split depending on the type of bond cleavage. Do the electrons stay together on one atom or do each of the electrons go back to their original atoms? In section R3.3.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/radicals-and-how-they-are-produced-id-43489/>), the concept of homolytic fission is introduced: when this bond cleavage occurs each atom receives one of the two electrons that formed the bond. Homolytic fission typically occurs in reactions involving free radicals.

What if the reaction involves charged species as it does in a nucleophilic substitution reaction (section R3.4.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>)))? This process of bond cleavage occurs differently as one atom receives both the electrons that formed the bond. How does it play a significant role in nucleophilic substitution reactions? How does heterolytic fission work compared to homolytic fission?

When the bond breaks apart and each atom gets one electron, this is homolytic fission. In heterolytic fission, one of the atoms will take both electrons leaving the other atom without any electrons.

Heterolytic fission causes the electrons of a bond to be allocated to one of the species forming two oppositely charged ions – one anion and one cation. Recall that when the covalent bond originally formed, each atom provided one electron. After heterolytic fission, the anion receives both electrons from the bond and therefore will have a negative charge since it contains one more electron than it did before the bond formed. The cation now has one less electron than it did

before the bond was formed, giving it a positive charge after heterolytic fissions (**Figure 1**). The electron pair movement in heterolytic fission is shown using the curly arrow, indicating the movement of a pair of electrons.



Figure 1. Homolytic and heterolytic fission of a covalent bond.

 More information for figure 1

The diagram illustrates two types of covalent bond fission: homolytic and heterolytic. In homolytic fission, denoted at the top, a bond between atoms X and Y splits evenly, with each atom retaining one electron, resulting in two free radicals ($\cdot\text{X}$ and $\cdot\text{Y}$). Curved arrows indicate electron movement. It highlights that this process creates radicals.

In heterolytic fission shown below, the bond between X and Y breaks unevenly, both electrons moving to atom Y. This produces a positively charged ion X^+ and a negatively charged ion $:\text{Y}^-$, represented by respective symbols and curly arrows demonstrating the electron pair movement. The process results in ion formation.

[Generated by AI]

Making connections

The way in which a bond breaks between two atoms is different in homolytic fission and heterolytic fission. Homolytic fission is covered in [subtopic R3.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43488/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43488/>), a pair of electrons of the bond between two atoms splits equally forming two free radicals that are highly reactive and will participate in chain reactions. When one of these radicals reacts with another molecule, it will produce a new radical.

On the other hand, the uneven splitting of electron pairs of a bond between two atoms in heterolytic fission results in the formation of a new covalent bond in a nucleophilic substitution reaction as it involves the departing of the leaving group. This will produce a new molecule and the ion of the leaving group.

In **Figure 2**, you can see that the breakage of covalent bonds of carbon and chlorine forms cation and anion. The electrons go towards the chlorine atom due to chlorine being more electronegative compared to carbon. This causes chlorine to have a partial negative charge and have a stronger pull on both electrons resulting in a carbocation and a chloride ion.



Figure 2. Heterolytic fission of the carbon—chlorine bond.

 More information for figure 2

The image is a diagram illustrating the process of heterolytic fission of a carbon—chlorine bond. On the left, a carbon (C) and a chlorine (Cl) atom are bonded together. The carbon atom is labeled with a delta positive (δ^+) sign, indicating a partial positive charge, and the chlorine atom is labeled with a delta negative (δ^-) sign, indicating a partial negative charge. Two pairs of dots are shown around the chlorine atom, representing lone pairs of electrons. An arrow curves from the bond towards the chlorine atom, symbolizing the movement of electrons during bond breakage.

To the right of the arrow, the carbon atom is depicted as a cation (C^+), having lost its shared electrons to chlorine, and the chlorine atom is shown as an anion (Cl^-), having gained those electrons. The diagram effectively conveys the shift of electrons towards the more electronegative chlorine resulting in ionic dissociation.

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The process of chlorine leaving chloropropane also can be explained using heterolytic fission. The chlorine atom on chloropropane has a partial negative charge and the carbon atom has a partial positive charge. The hydroxide ion as a nucleophile is attracted to the partially positively charged carbon atom and attacks the carbon resulting in the breaking of the bond between carbon and chlorine. The bonding electrons between these two atoms are transferred to the chlorine atom and cause it to detach from the substrate forming a chloride ion, known as the leaving group.

Figure 3 shows another example of heterolytic fission.



Figure 3. Heterolytic fission of the carbon—hydrogen bond.

 More information for figure 3

The image is a diagram illustrating the heterolytic fission of a carbon-hydrogen bond. On the left, there is a representation of a bond between a carbon (C) atom and a hydrogen (H) atom. The carbon atom has a partial negative charge (δ^-), and the hydrogen atom has a partial positive charge (δ^+). An arrow indicates the direction of electron movement from hydrogen to carbon. On the right side, the resulting ions after the bond breakage are shown: a carbon ion with a full negative charge (C^-) and a hydrogen ion with a full positive charge (H^+). This depicts the shift of bonding electrons to the carbon atom upon bond cleavage, aligning with the difference in electronegativity between carbon and hydrogen.

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Referring to [section S2.2.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) and the electronegativity data that are given in [section 9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet, the electronegative value of hydrogen is smaller than carbon. Therefore, carbon will have a partial negative charge as it is more electronegative, and hydrogen will have a partial positive charge. When the bond breaks, the electrons will go towards the carbon atom resulting in carbon with a negative charge and hydrogen with a positive charge.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication — Collaborate with peers to share ideas, listen actively to other perspectives and encourage others to contribute
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Materials

- modelling clay, at least four different colours, preferably: black for carbon, white for hydrogen, orange for bromine, green for chlorine and red for oxygen
- toothpicks
- paper.

Your task

1. Create two atomic models of the same molecule that contains a bond that will break through homolytic and heterolytic fission.
2. Round up the plasticine like a ball as this will represent the atoms and connect them with toothpicks. When creating the molecules, you should ensure that it contains at least carbon and hydrogen atoms.
3. Choose one person in your group to break one of the bonds on one molecule that can undergo a homolytic fission and ask another person in your group to break another molecule that can undergo a heterolytic fission.
4. Set the broken toothpicks and molecules/atoms such as in **Figure 3** (molecules before and after bond cleavage).
5. Cut the paper to make the curly arrows and place it correctly on both bond cleavages. Round up the plasticine into small balls and place them to show electrons.

Discuss whether your fellow group member has broken the correct bonds on both molecules. Are the bond cleavages set up correctly? Is the direction of the curly arrow correct? How are these two bond cleavages involved in bond breaking of an electron pair? How do these two bond cleavages affect the bond formation? What kind of reactions of the products would form these two bond cleavages?

After that, you can ask another person in your group to show the molecule that has been made and repeat the same process.

Electrophilic addition reaction

Learning outcomes

At the end of this section you should be able to:

- (R3.4.4) Recognise the electrophile in chemical reactions.
- (R3.4.5) Deduce equations for the reactions of alkenes with water, halogens and hydrogen halides.

The formation of a new covalent bond occurs when nucleophiles attacks electrophiles. This type of reaction is very common in carbon-based reactions such as substitution and addition reactions. How does an addition reaction differ from a substitution reaction? The term 'addition reaction' is explained in [section R3.2.11](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/>) describing a hydrogen atom that is added to reduce the degree of an unsaturated compound such as alkene or alkynes. How do you identify the electrophile in a chemical reaction?

Electrophiles in chemical reactions

Figure 4 in the [previous section](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/heterolytic-fission-id-46328/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/heterolytic-fission-id-46328/>) shows the hydroxide ion attacking the partially positively charged carbon atom in bromoethane. This type of electron-deficient carbon is called an electrophile. This carbon atom is looking for an electron-rich species to fill its electron-deficient site by forming a new covalent bond. Electrophile means an electron-deficient species (neutral or positively charged atom) and it has a tendency to accept or to attract a pair of electrons from the high electron density region.

What are some common types of electrophile and what types of molecule do they usually react with?

An electrophile has an electrophilic centre atom, often containing an incomplete octet, which carries a partial or full positive charge. These can be protons, carbocations, halogen molecules or hydrogen halides, any molecule that has a carbonyl group and molecules with incomplete octets such as boron trifluoride.

The electrophile as a deficient-electron species seeks an electron pair from the nucleophiles to complete its valence shell. The reaction between a nucleophile and an electrophile occurs in electrophilic addition or substitution reactions. Common examples of electrophiles are shown in **Figure 1**.

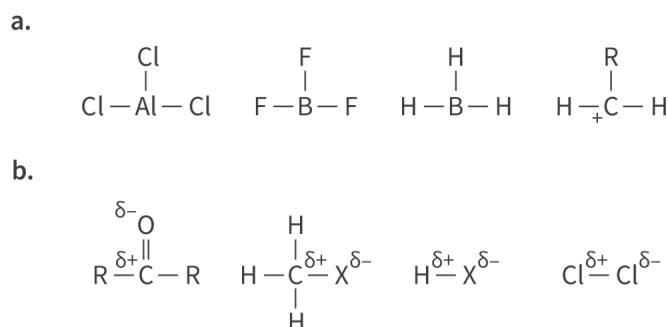


Figure 1. Examples of electrophiles. **a.** Contain atoms with an incomplete octet. **b.** Contain partial positively charge atom.

 More information for figure 1

The diagram shows examples of electrophiles, divided into two parts labeled 'a.' and 'b.'

Part 'a.' illustrates chemical structures containing atoms with an incomplete octet: - Cl—Al—Cl, a linear structure with chlorine atoms bonded to aluminum. - F—B—F, a structure with fluorine atoms bonded to boron. - H—B—H, a similar structure with hydrogen atoms bonded to boron. - H—C⁺—H, where carbon is positively charged, bonded to two hydrogen atoms.

Part 'b.' displays structures containing partially positively charged atoms: - R—C(δ^+)—O(δ^-)=C—R, illustrating the carbon and oxygen atoms with delta positive and negative charges, respectively. - H—C(δ^+)—X(δ^-), showing hydrogen, carbon, and a generic atom X with partial charges. - Cl(δ^+)—Cl(δ^-), demonstrating chlorine atoms with partial charges. - H—X(δ^+)—X(δ^-), showing hydrogen bonded to a generic atom X with partial charges.

[Generated by AI]

An electrophile is a species that can not only accept a lone pair from the nucleophile, but can also initiate the electron movement by attracting the electron-rich species towards itself. The attraction between the electrophile and nucleophile causes the transfer of electrons to occur, leading to the formation of new bonds.

For example, the reaction between ethene and hydrogen bromide (**Figure 2**).



Hydrogen bromide as an electrophile is attracted to the electron-rich double bond of ethene, causing the bond between the two carbon atoms to break. This opens up two new bonding positions, one on each carbon atom. The bond between the H and Br in HBr undergoes heterolytic bond fission. The hydrogen attaches to one of the carbon atoms and the bromide ion attaches to the other carbon atom, producing 1-bromoethane.

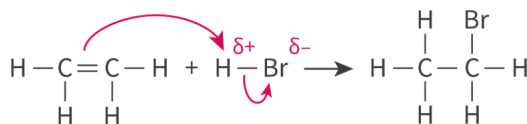


Figure 2. Electrophiles initiate electron movement in addition reactions.

 More information for figure 2

The diagram illustrates the electrophilic addition reaction of hydrogen bromide (HBr) to ethene, forming 1-bromoethane. On the left, the structure of ethene is shown with two carbon atoms double-bonded and each carbon having two hydrogen atoms attached. A curved arrow indicates the movement of electrons from the double bond towards the hydrogen atom in HBr. Additionally, there is a delta positive (δ^+) symbol near the hydrogen and a delta negative (δ^-) symbol near the bromine atom, signifying polarity in HBr.

As the reaction progresses, the bond between the carbon atoms in ethene breaks, and a new bond forms between one of the carbon atoms and the hydrogen (H) from HBr. This results in the other carbon atom forming a bond with the bromide (Br) ion. The final structure on the right shows 1-bromoethane, where one carbon atom is bonded to three hydrogen atoms and the other carbon is bonded to bromine and two hydrogen atoms. The diagram captures the key steps and electron movement involved in electrophilic addition reactions.

[Generated by AI]

Figure 3 shows an example of how the nucleophile attacks an electrophile. The hydroxide ion (the nucleophile), contains a non-bonding pair of electrons, collides with the carbon atom in carbocation that is positively charged (as an electrophile). The negative electrons are strongly attracted to the positively charged carbon, which then accepts the pair of electrons from the hydroxide ion, forming a coordination bond. Other examples of an electrophile in chemical reactions are shown in **Figure 4**, where the carbon in bromoethane obtains electrons from the hydroxide ion.

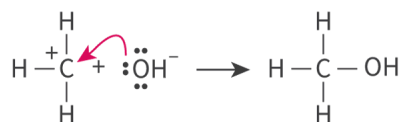


Figure 3. A nucleophile attacks an electrophile.

 More information for figure 3

The image depicts a chemical reaction where a hydroxide ion (OH^-) acts as a nucleophile and attacks a carbocation (C with a positive charge) which is acting as an electrophile. The diagram illustrates the interaction between the negatively charged hydroxide ion, which has a pair of non-bonding electrons, and the positively charged carbon atom in the carbocation. An arrow points from the electrons on the hydroxide ion to the carbon atom, indicating the movement of electrons. As a result, the carbon atom accepts the pair of electrons and forms a new coordination bond with the hydroxide ion, turning into a neutral alcohol molecule, with the hydroxide now bonded to the carbon. The diagram highlights the transformation from a carbocation to an alcohol.

[Generated by AI]

Figure 4 is another example of a neutral electrophile receiving a pair of electrons from 2-bromo-2-methylpropane and forming a new covalent bond. The bond that is formed between bromine and aluminium atoms is also called a coordination bond where the electrons are shared from one atom. The coordination bond is introduced in section S2.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>) and is explained in more detail in the HL section R3.4.7–8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/>).

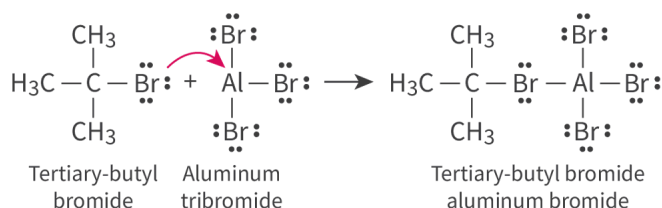


Figure 4. A nucleophile attacks a neutral electrophile.

 More information for figure 4

The diagram shows a chemical reaction illustrating a nucleophilic attack. On the left, there is a structure labeled "Tertiary-butyl bromide" showing a carbon atom connected to three methyl groups (CH_3) and a bromine atom (Br) with lone pairs. This structure is interacting with another compound labeled "Aluminum tribromide," consisting of an aluminum atom connected to three bromine atoms with lone pairs. An arrow indicates the movement of electrons from the bromine atom in the tertiary-butyl bromide towards the aluminum atom in aluminum tribromide. On the right of the arrow, a new structure is formed, labeled "Tertiary-butyl bromide aluminum bromide," showing the tertiary-butyl group now connected to an aluminum-bromine complex through a coordinate bond. This bond is formed as a result of electron sharing from the bromine atom.

[Generated by AI]

**Exercise 1**

Click a question to answer



1. Which of the following statements is true about an electrophile in electrophilic addition reactions?


☒ Check answers

Reactions of alkenes with water, halogens and hydrogen halides

An electrophile is also involved in addition reactions by attacking the unsaturated bond of unsaturated organic compounds such as alkenes and forming a new compound. The structure of alkenes ensures these molecules have a high electron density due to their double bond because this is susceptible to electrophilic attack. The general addition reaction in an alkene is illustrated in **Figure 5**. Here, 'R' represents either a hydrogen or alkyl group and 'A-B' represents a molecule that is being added across the double bond and forms a single bond.

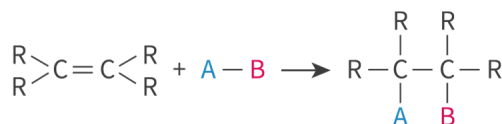


Figure 5. Electrophilic addition reaction of alkenes.

More information for figure 5

The image is a chemical diagram depicting an electrophilic addition reaction of alkenes. It shows a structural formula where a molecule represented as "R"-substituted alkene with a carbon-carbon double bond ($\text{R}-\text{C}=\text{C}-\text{R}$) reacts with a molecule "A-B". After the reaction, the result is a new molecule where "A" and "B" have been added across the former double bond, leading to a single bond structure: $\text{R}-\text{C}(\text{A})-\text{C}(\text{B})-\text{R}$. The text in the diagram uses different colors for "A" and "B" to differentiate them from other elements in the structure, indicating their addition across the double bond.

[Generated by AI]

Alkenes are more reactive compared to alkanes because these unsaturated hydrocarbons contain at least one double bond between the two carbon atoms, and this is where addition reactions such as halogenation, hydrohalogenation and hydration (**Figure 5**) take place. An example is the reaction between ethene and bromine producing 1,2-dibromoethane. The addition reaction in which alkenes are combined with bromine water, $\text{Br}_2(\text{aq})$, is used to test the presence of a double bond in hydrocarbons. This reaction occurs at room temperature quickly.

Making connections

Alkanes contain only a single bond (see [section S3.2.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>)), making them unreactive towards most chemical reagents such as bromine. Bromine water is orange in colour, and will stay the same when it is mixed with an alkane, but the orange colour fades if its molecules react with the double bonds in alkenes. This colourless solution contains di-halogen alkane that is produced as the double bond is broken. The reaction between alkenes and bromine water is used to differentiate between alkanes and alkenes, and is known as the bromine water test. It is commonly used to identify the presence of double bonds in hydrocarbon samples.

The electrophilic addition of alkenes also occurs with the addition of halogen halide (HX) such as when ethene reacts with hydrogen chloride producing chloroethane (**Figure 5**). The reaction will occur through a concentrated solution of hydrogen halide. Another example of electrophilic addition is hydration, in which alkenes are converted to alcohol by adding water (**Table 1**). Concentrated sulfuric acid is often used to speed up this reaction. This type of reaction is essential as it can be used to produce ethanol on a large scale.

Table 1. Addition reaction examples.

Reactions	Chemical equations
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{Br}-\text{Br} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Bromine} \quad \text{1,2-dibromoethane} \\ \text{Halogenation} \end{array} $ More information	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{Br}_2(\text{aq}) \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}(\text{aq}) \end{array} $
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{H}-\text{Cl} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Hydrogen chloride} \quad \text{1-chloroethane} \\ \text{Hydrohalogenation} \end{array} $ More information	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{aq}) \rightarrow \text{C}_2\text{H}_5\text{Cl}(\text{aq}) \end{array} $
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{H}-\text{OH} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Water} \quad \text{Ethanol} \\ \text{Hydration} \end{array} $	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \end{array} $

Reactions	Chemical equations
👁 More information	$\text{C}_2\text{H}_5\text{OH}(\text{l})$

The reaction between alkenes and HI is more rapid than with HCl. HI is a stronger acid than HCl, and dissociates more readily producing H^+ and I^- ions in solution (see [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>)). Also, HCl has a stronger bond (see [section S2.2.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>)) that makes it less ready to react with alkenes. Therefore, the reaction of alkenes and HCl often requires catalysts such as AlCl_3 .

Strand: Service

Learning outcome: Demonstrate how to initiate and plan a CAS experience.

How healthy is your diet? A balanced diet is essential not only for your physical but also mental well-being. In recent years, the subject of trans fats  (https://medlineplus.gov/ency/patientinstructions/000786.htm#:~:text=Trans%20fats%20are%20made%20when,from%20vegetable%20oil%20in%20a%20reaction%20known%20as%20hydrogenation.) has been hitting the headlines. Trans fats are produced when hydrogen is added to vegetable oil in a reaction known as hydrogenation. In 2015, the Food and Drug Administration  (https://www.fda.gov/food/food-additives-petitions/trans-fat#:~:text=Most%20of%20the%20trans%20fat,This%20process%20is%20called%20hydrogenation.) (FDA) classified trans fats as being no longer generally recognised as safe to eat due to the negative health effects associated with eating them.	
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One possible CAS experience could be to research the guidelines or regulations about the use of trans fats in your country or in a country of your choice. You could also research whether imported foods are subject to the same legislation as locally produced food.

You can check your understanding of addition reactions in alkenes in the activity below.

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Your task

There are seven reactions in **Interactive 1** and they are halogenation, hydrohalogenation and hydration of alkene reactions. Arrange the reactants including their reagents and products with a partner by dragging and dropping the structures and reagents into the correct places.

With practise, you will be able to quickly predict the products or reactants of the addition reaction of alkenes.

Reactions	Reactants	Reagent	Products
Halogenation of propene			
Hydrohalogenation of but-1-ene			
Hydration of 2-methylpropene			
Hydrohalogenation of 2-methylpent-2-ene			

$\xrightarrow{\text{H}_2\text{O}}$	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{C}=\text{CH}_2 \\ \\ \text{H}_3\text{C} \end{array}$	$\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_3$	$\begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
$\begin{array}{c} \text{Br} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	$\xrightarrow{\text{Br}_2}$	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \\ \text{Cl} \end{array}$	$\xrightarrow{\text{HBr}}$
$\text{H}_2\text{C}=\text{CH}-\text{CH}_3$	$\begin{array}{c} \text{H}_2\text{C}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{Br} \quad \text{Br} \end{array}$	$\xrightarrow{\text{HCl}}$	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{C}=\text{CH} \\ \quad \\ \text{H}_3\text{C} \quad \text{CH}_2-\text{CH}_3 \end{array}$

☒ Check

Interactive 1. Arrange the reactants including their reagents and products.

Discuss with your partner the challenges you encounter when arranging these organic compounds. How does this activity help you to understand this section?

Extension task

Use what you learned in [section S2.2.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>) to name all the products formed from each reaction.

Lewis acids and bases (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to:

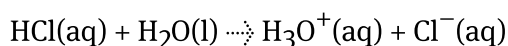
- Apply Lewis acid–base theory to inorganic and organic chemistry to identify the role of the reacting species.
- Explain the relationship between Brønsted–Lowry acids and bases and Lewis acids and bases.

Section R3.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/>) covers the Brønsted–Lowry acid and base interaction in which a proton must be transferred. This theory works well for common acids and bases in which molecules accept and donate protons. However, this theory does not hold for all acid–base interactions. One example is the identification of iron(III) ions as acidic in water. How can these ions show acidity if they do not have any protons to donate? What about the participation of electron pairs in determining the acidic and basic behaviour of molecules? Electron pairs play an important role because when a molecule donates or accepts protons, it leaves behind or gains an electron pair.

These exceptions can be explained by the Lewis acids and bases theory, which focuses on electron transfer rather than proton transfer. This concept is more general, so it can cover most substances. The interaction of electron pairs plays a significant role in identifying both a Lewis acid and Lewis base during a reaction. What is the role of a lone pair in Lewis acids and bases? How do you identify them? Is there any relationship between Brønsted–Lowry acids and bases and Lewis acids and bases?

Definition of a Lewis acid and base

Before discussing the definition of Lewis acids and bases, we will review the concept of Brønsted–Lowry acids and bases, in which a proton donor substance is called an acid and a proton acceptor substance is called a base. For example, the reaction between hydrochloric acid and water.



This reaction shows that a proton is donated by the hydrochloric acid as an acid to water, which acts as the base forming a hydronium ion.

Now, think about how this proton transfers from the perspective of the electron movement for the breaking and forming of covalent bonds (**Figure 1**). This process also involves transfer of an electron as the partial negative charge oxygen atom in water attracts the partial positively charged hydrogen atom in hydrogen chloride, causing the covalent bond in this molecule to break. As a result, the hydrogen atom as a proton forms a new bond with the oxygen atom, forming a hydronium ion. The chlorine atom is then left with an extra electron that becomes a chloride ion.

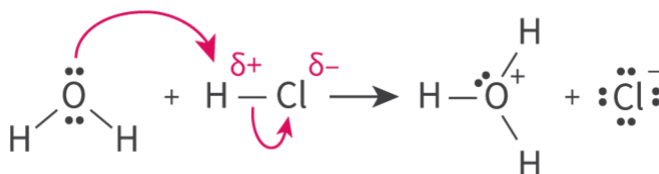


Figure 1. Proton transfer mechanism.

 More information for figure 1

The diagram illustrates the mechanism of proton transfer in a chemical reaction between water (H_2O) and hydrogen chloride (HCl). On the left, a water molecule (H_2O) is shown with two hydrogen atoms and one oxygen atom, which has two lone pairs of electrons. The oxygen has an overall neutral charge.

In the middle, an HCl molecule is depicted, with a partial positive charge (δ^+) on the hydrogen and a partial negative charge (δ^-) on the chlorine, indicating a polar covalent bond.

The diagram uses curved arrows to indicate the movement of electrons. A curved arrow goes from one of the lone pairs on the oxygen in water to the hydrogen in HCl , showing that the oxygen donates an electron pair to form a bond with the hydrogen, creating a hydronium ion (H_3O^+) on the right.

A parallel arrow shows the breaking of the bond between H and Cl in HCl , transferring an electron to the chlorine atom, resulting in a chloride ion (Cl^-), which is depicted with an extra pair of electrons.

[Generated by AI]

This process shows how electron transfer is crucial in chemical reactions, which leads to the Lewis acid–base theory based on electron-pair transfer. The interaction of electrons from reactant molecules leads to formation of a new chemical as a pair of electrons is donated by Lewis base and received by the Lewis acid (**Figure 2**). This theory defines acid as a substance that can accept a lone pair and base is a substance that can donate a lone pair.

Common examples of Lewis acids are protons (H^+ ion), cations of metals such as magnesium ion, cations of d-block elements such as a silver ion, atoms that have incomplete valence shells (such as aluminium and boron) and carbocation and

carbonyl compounds.



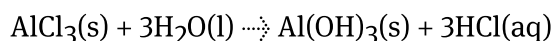
Figure 2. Formation of coordination bond by the interaction between a Lewis acid (A) and a Lewis base (B).

 More information for figure 2

The diagram illustrates the formation of a coordination bond through the interaction of a Lewis acid and a Lewis base. On the left, the letter 'A' is labeled as a Lewis acid, and to its right, ':B' is labeled as a Lewis base. An arrow extends from 'A' to ':B', indicating the donation of an electron pair by the Lewis base to the Lewis acid. The result of this interaction is shown to the right with 'A - B', representing the newly formed coordination bond.

[Generated by AI]

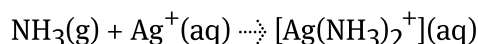
For example, AlCl_3 has an incomplete valence shell although it shares its three electrons with chlorine atoms and forms three single bonds. Therefore, it is still able to receive electron pairs and this is a behaviour of Lewis acid.



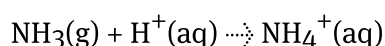
On the other hand, the Lewis base is commonly characterised as an electron-rich or negatively charged atom. The electron pairs that are accepted by the Lewis acids from Lewis bases are used to form a covalent bond. Common examples of Lewis bases are water, ammonia, hydroxide ions, cyanide ions, halide ions and ethanoate ions.

Lewis acid—base theory with inorganic and organic examples

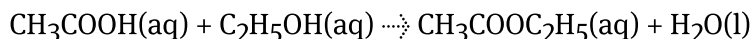
Looking at the previous examples, this Lewis acid–base theory can be applied to both organic and inorganic chemistry. In inorganic chemistry, metal cations commonly act as Lewis acids due to their positive charge. For example, ammonia acts as a Lewis base and donates a pair of electrons to the silver ion or hydrogen ion as a Lewis base.



or



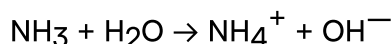
Whereas, in organic chemistry, carbocation or carbonyl compounds often act as Lewis acids as they have an electron-deficient carbon atom that accepts an electron pair from nucleophiles such as hydroxide ions or alcohol. This occurs in esterification reactions where an ester is formed when an alcohol is reacted with a carboxylic acid.



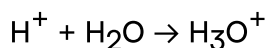
Study skills

Water can be both identified as a Lewis acid or a Lewis base, depending on the reaction it is involved in. Therefore, water is also said to be amphoteric (see [subtopic R3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/)).

It acts as a Lewis acid and receives a pair of electrons from ammonia.



It acts as a Lewis base and donates a pair of electrons to hydrogen ions.



The relationship between Brønsted—Lowry and Lewis acid—base theory

In comparison with the Brønsted—Lowry acids and bases (see [section R3.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/)), Lewis acids and bases theory is broader as it is not restricted to just those reactions that include the transfer of protons or water as solvents.

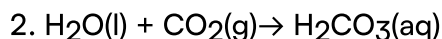
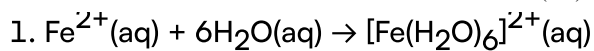
In fact, all Brønsted—Lowry acids are also considered Lewis acids because of the availability to donate protons and therefore are also able to accept an electron pair. On the other hand, not all Lewis acids are Brønsted—Lowry acids. For example, a transition metal ion in a complex ion can act as a Lewis acid by accepting an electron pair, but it cannot act as a Brønsted—Lowry acid, as it cannot donate a proton.

Similarly, all Brønsted—Lowry bases are considered Lewis bases as all Brønsted—Lowry bases have a lone pair to accept hydrogen and all Lewis bases have lone pairs. Conversely, all Lewis bases are not Brønsted—Lowry bases as Lewis bases can act as nucleophiles, rather than Brønsted—Lowry bases.

Try **Worked example 1** and the exercise question on the Lewis acid—base theory.

Worked example 1

Identify and explain the role of each reactant in these reactions according to the Lewis acid—base theory.



Lewis acids: _____

Lewis bases: _____

Lewis acids: 1. Fe^{2+} and 2. CO_2

Lewis bases: 1. and 2. H_2O

The iron(II) ion and carbon dioxide act as a Lewis acid as they accept an electron pair from water as a Lewis base, forming a new bond.

Activity

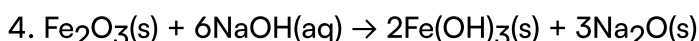
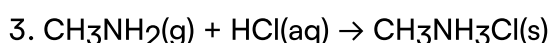
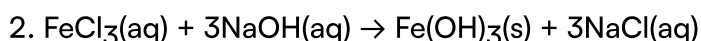
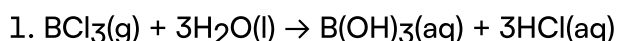
- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
 - Social skills — Working collaboratively to achieve a common goal
 - Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

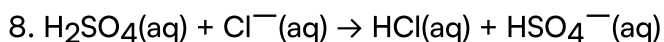
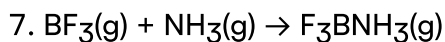
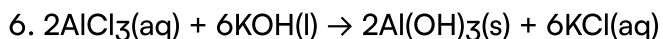
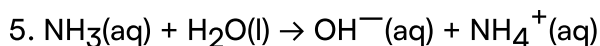
Your task

With your partner, identify the reactants in the reactions below and state whether each one can be categorised as a Lewis acid, Lewis base, Brønsted—Lowry acid or Brønsted—Lowry base. Then identify Lewis acids that are not Brønsted—Lowry acids and Lewis bases that are not Brønsted—Lowry bases from the same reactions, and complete the table provided in the worksheet.

Worksheet 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry%20Worksheet%20Lewis%20Acids%20and%20Bases.pdf)





Discuss and draw a conclusion with your partner as to what the similarities and differences are between Lewis acid—base theory and Brønsted—Lowry acid—base theory. Also, discuss the differences and similarities between Lewis acids and Brønsted—Lowry acids, and between Lewis bases and Brønsted—Lowry bases.

Brønsted—Lowry acids and Lewis acids

In reaction 1, BCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 2, FeCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 3, HCl is a Lewis acid and a Brønsted—Lowry acid.

In reaction 4, Fe_2O_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 5, H_2O is a Lewis acid and a Brønsted—Lowry acid.

In reaction 6, AlCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 7, BF_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 8, H_2SO_4 is a Lewis acid and a Brønsted—Lowry acid.

Brønsted—Lowry bases and Lewis bases

In reaction 1, H_2O is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 2, NaOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 3, CH_3NH_2 is a Lewis base and a Brønsted—Lowry base.

In reaction 4, NaOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 5, H_2O is a Lewis base and a Brønsted—Lowry base.

In reaction 6, KOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 7, NH_3 is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 8, Cl^- is a Lewis base and a Brønsted—Lowry base.

Coordination complex (HL)

Higher level (HL)

Learning outcomes

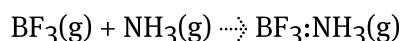
At the end of this section, you should be able to:

- (R3.4.7) Draw and interpret Lewis formulas of reactants and products to show coordination bond formation in Lewis acid—base reactions.
- (R3.4.8) Deduce the charge on a complex ion, given the formula of the ion and ligands present.

In [section S2.2.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>), a Lewis formula is used to show all valence electrons involved representing the covalent bonding. This formula also can be used to show the bond that is formed due to the shared pair of electrons from the same atom, known as a coordination bond (see [section S2.2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>)). This bond can also be formed between a metal ion and ligand by sharing of a pair of electrons, which often forms brightly coloured compounds. For instance, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ forms a bright blue compound. The concept of sharing a pair of electrons is also connected with the previous [section R3.4.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>), in which nucleophiles donate a pair of electrons, which is also called Lewis base (see later in this section). What is the relationship between a coordination bond with a nucleophile and Lewis base? What is their connection?

Lewis acid—base interaction: formation of coordination bond

The interaction between a Lewis acid and Lewis base results in a formation of a coordination bond as an electron pair that is donated from the Lewis base coming from the same atom. This typical bond is always produced in a Lewis acid–base reaction. For example, ammonia donates a lone pair as Lewis base to boron trifluoride as Lewis acid because it has an incomplete octet (**Figure 1**).



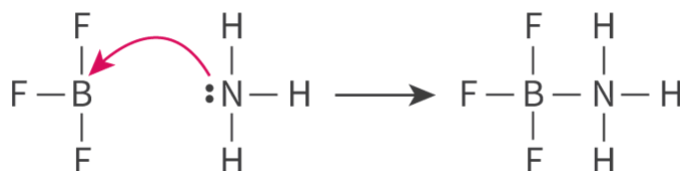


Figure 1. A lone pair from ammonia is donated to boron trifluoride.

 More information for figure 1

The image is a chemical reaction diagram illustrating the interaction between ammonia (NH₃) and boron trifluoride (BF₃). On the left side, the structure of BF₃ is shown with boron (B) at the center and three fluorine (F) atoms bonded to it. Ammonia (NH₃) is depicted with nitrogen (N) at the center, bonded to three hydrogen (H) atoms. A lone pair of electrons is represented as a pair of dots adjacent to the nitrogen. An arrow points from the lone pair to the boron atom, indicating the donation of electrons to form a coordinate bond. On the right side, the resulting compound is displayed as F-B-N-H, with B now having a coordinate bond to N, signifying the formation of a new adduct.

[Generated by AI]

The nitrogen atom (in ammonia) donates a pair of the non-bonding electrons to a boron atom that has an incomplete valence shell. What does this process have in common with nucleophilic attack?

A nucleophile is attracted to an electrophile and donates an electron pair forming a new covalent bond (see [section R3.4.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>)). This process is similar to the Lewis acid–base interaction in which the formation of a new bond due to an electron pair is donated by the Lewis base. However, the bond that is formed due to this interaction is called a coordination bond.

Making connections

The properties of coordination bonds compared to other covalent bonds are similar.

Once the coordination bond is formed, it is indistinguishable from other covalent bonds. The only difference is the way in which the bond is formed. The coordination bonds are formed by sharing a pair of electrons from the same atom (see [section R3.2.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/print/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/print/>)).

[753301/book/oxidation-and-reduction-id-43487/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/)), whereas the bonds of other covalent bonds are formed by electrons that are shared equally from the two atoms (see [section S2.2.1a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>)).

Therefore, this is similar to the process that occurs between the nucleophilic attack and Lewis acid–base interaction involving donation and acceptance of an electron pair. In other words, a Lewis base is also called a nucleophile and a Lewis acid is an electrophile. Another example of the formation of coordination bonds is illustrated in **Figure 2** in which an electron pair from nitrogen is donated to hydrogen ions.

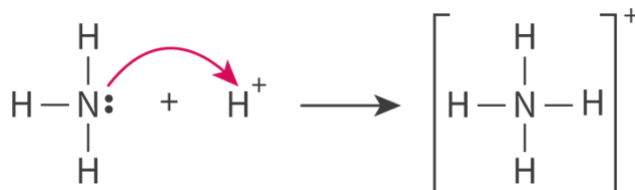


Figure 2. Ammonia protonation.

 More information for figure 2

The image is a chemical diagram illustrating ammonia protonation. It depicts a structural formula where nitrogen (N) is bonded to three hydrogen (H) atoms, presented with a lone pair represented by a pair of dots next to nitrogen. An arrow points from the lone pair to a hydrogen ion (H⁺), indicating the donation of the electron pair from nitrogen to the hydrogen ion. This forms a new bond, shown on the right part of the diagram, where the nitrogen is now positively charged and forms a coordinate bond with the additional hydrogen, resulting in an ammonium ion (NH₄⁺). The ammonium ion is enclosed in square brackets with a positive charge indicating its formation.

[Generated by AI]

The charge on a complex ion

In [section 3.1.8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>), transition elements are described as having incomplete d-sublevels that give them characteristic properties such as the formation of complex ions with ligands. For example, $[\text{Fe}(\text{CN})_6]^{4-}$ naturally forms bonds between ligands and the transition element. How does this relate with the Lewis acid–base interaction? Which is acting as the Lewis acid and which is acting as the Lewis base?

Iron has unoccupied d-orbitals accepting the non-bonding pair of electrons from the cyanide ion, CN^- . As it involves acceptance and donation of an electron pair, the formation of ion complex is related to

Lewis acid—base interaction. Iron is identified as a Lewis acid and cyanide ion is identified as a Lewis base.

Another example is the formation of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ions. Copper(II) ions (Cu^{2+}) in aqueous solution receive an electron pair from water as a Lewis base or ligand (**Figure 3**). This is because copper(II) is surrounded by water molecules as ligands.

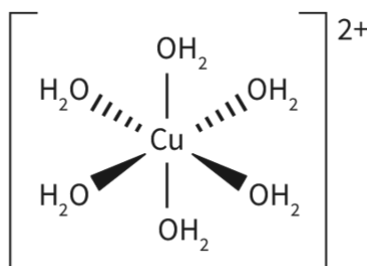
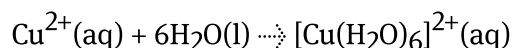


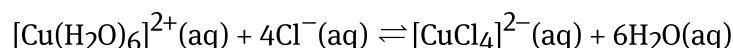
Figure 3. The $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ion.

 More information for figure 3

This is a structural diagram of the copper hexaaqua complex ion, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$. At the center of the diagram is a copper ion (Cu). Surrounding the central Cu are six water molecules, each denoted as H_2O . These water molecules are arranged in an octahedral geometry around the copper ion. The bonds between the copper ion and the water molecules are depicted with different lines: solid lines for in-plane bonds, dashed lines for bonds extending outwards, and bold lines for bonds going inwards. This geometric arrangement illustrates the spatial orientation of the water molecules relative to the central copper ion, showcasing the characteristic structure of the complex ion.

[Generated by AI]

Complex ions may experience a substitution reaction such as water in aqueous solution being replaced by other ligands.



The number of coordination bonds formed by central metal ions and ligands is called the coordination number. The strength and stability of the coordination bond in complex ions is affected by the net charge of both ligands and the central metal ion. The electrostatic force will be strong in complex ions if both the opposite charges from ligands and the central metal ion are high. Neutral ligands such as H_2O , NH_3 and CO have a charge of zero, and negatively charged ligands such as Cl^{-} ,

CN^- and Br^- have a charge of minus one (-1). To help you understand how to deduce the charge on a complex ion, **Figure 4** shows the components of one in detail.

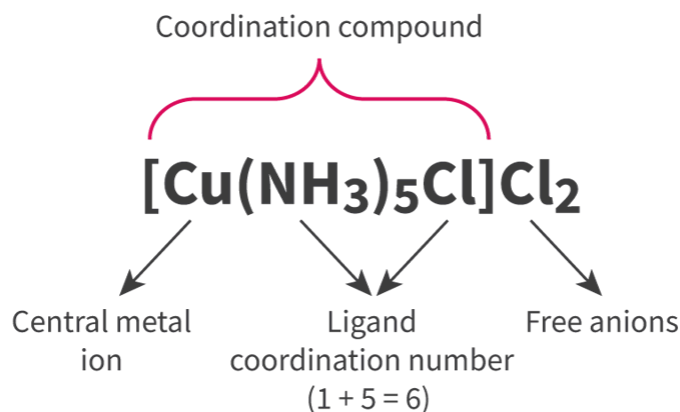


Figure 4. The formula of ions and ligands in a complex ion.

 More information for figure 4

This diagram illustrates the formula of a complex ion, specifically focusing on the coordination compound. At the center of the diagram is the formula $[\text{Cu}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, which represents a copper complex ion. Above the formula, there is a label "Coordination compound" with a bracket indicating that the entire formula represents the coordination compound. Arrows point to different parts of the formula, providing additional information. The label "Central metal ion" points to Cu, indicating that copper is the central metal ion. "Ligand coordination number (1 + 5 = 6)" points towards $(\text{NH}_3)_5$ and Cl within the brackets, specifying that there are six ligands coordinated to the central metal ion. Lastly, the label "Free anions" points to Cl_2 outside the bracket, indicating that these are free anions. This visual breaks down the components of a complex ion, highlighting the relationships and elements involved in such structures.

[Generated by AI]

Try the worked examples and the exercise question on charges on complex ions.

Worked example 1

Cobalt(III) can form a complex ion with four ammonia and two chloride ligands. Calculate the overall charge and formula of the complex ion.

Charge of cobalt(III) is 3+

Charge of ammonia as a neutral ligand is 0

Charge of chloride is 1-

The overall charge is $(+3) + 4(0) + 2(-1) = 1+$

The formula of the complex ion is $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$

Worked example 2

Deduce the charge of chromium in $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$.

The overall charge is neutral, so:

$$\begin{aligned}(\text{Cr}) + 6(\text{H}_2\text{O}) + 3(\text{Cl}) &= 0 \\(\text{Cr}) + 6(0) + 3(-1) &= 0 \\ \text{Cr} &= 3+\end{aligned}$$



Exercise 1

Click a question to answer



1. A complex ion of $[\text{CrCl}_2(\text{NH}_3)_4]^+$ has chromium as a central metal ion with the charge of _____, and chlorine as well as ammonia as ligands.

Check answers

Higher level (HL)

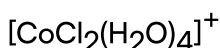
Activity

- IB learner profile attribute: Communicator
- Approaches to learning:

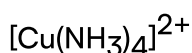
- Communication skills — Using digital media for communicating information
- Research skills — Evaluating information sources for accuracy, bias, credibility and relevance
- Time required to complete activity: 30—40 minutes
- Activity type: Group activity

Your task

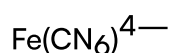
Form a group of three. Each person researches one of the three complex ions:



or



or



1. After you have selected one of these three complex ions, research the application of your complex ion in industry.
2. Next, create a digital animation showcasing the structure of your chosen complex ion. This animation should accurately display the molecular geometry and the location of the lone pairs. There are various websites available you can use to create your videos or digital animations (e.g. [Flipgrid](https://info.flip.com/)  (<https://info.flip.com/>), [Animaker](https://www.animaker.com/)  (<https://www.animaker.com/>)).

Work collaboratively with your group to check that the information you give is accurate, such as the number of lone pairs of the ligands and the structure of the complex ion. Discuss any further suggestions and check for false information in your explanation to ensure everyone has a chance to participate in the evaluation process.

Nucleophilic substitution reaction mechanism (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.4.9) Describe and explain the mechanisms of the reactions of primary and tertiary halogenoalkanes with nucleophiles.
- (R3.4.9) Explain the stereospecific nature of S_N2 .
- (R3.4.10) Predict and explain the relative rates of the substitution reactions for different halogenoalkanes (RCl, RBr, RI).

Section R3.4.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>) describes halogenoalkanes undergoing substitution reactions in which a halogen atom is replaced by a nucleophile. This can be illustrated by an orange cat replacing a grey cat that was sleeping in a bed. The grey cat represents a halogen atom in a halogenoalkane compound. The halogenoalkane is a molecule with a carbon bonded to a halogen atom like a chain with one cat at one end. Imagine there is a sleepy grey cat (halide ion) comfortable in her spot, enjoying a peaceful nap. The energetic and playful orange cat (nucleophile) is eager to explore and spots the bed with the grey cat in it. The orange cat is intrigued, sneaks up and quietly approaches the bed out of curiosity. With a sudden burst of energy, the orange cat is kicking out and jumps into the bed, startling the grey cat. The orange cat settles into the bed, and the grey cat is now displaced from the comfortable spot. The orange cat has replaced the grey cat in the bed. The halogen atom is 'kicked out' or replaced by the nucleophile.

Nucleophilic substitution reactions can occur through two possible mechanisms. Imagine that you have two different halogenoalkanes, for example chloromethane (CH_3Cl) and 2-chloro-2-methylpropane ($(\text{CH}_3)_3\text{CCl}$), reacting with hydroxide ions to produce an alcohol (**Figure 1**). The characteristics of the products are different as the reaction occurs through different substitution reaction mechanisms. What factors might influence the outcome of the reaction for both halogenoalkanes? Is it related to the structure of the halogenoalkanes or the reaction conditions? What are the differences between the two reaction mechanisms?

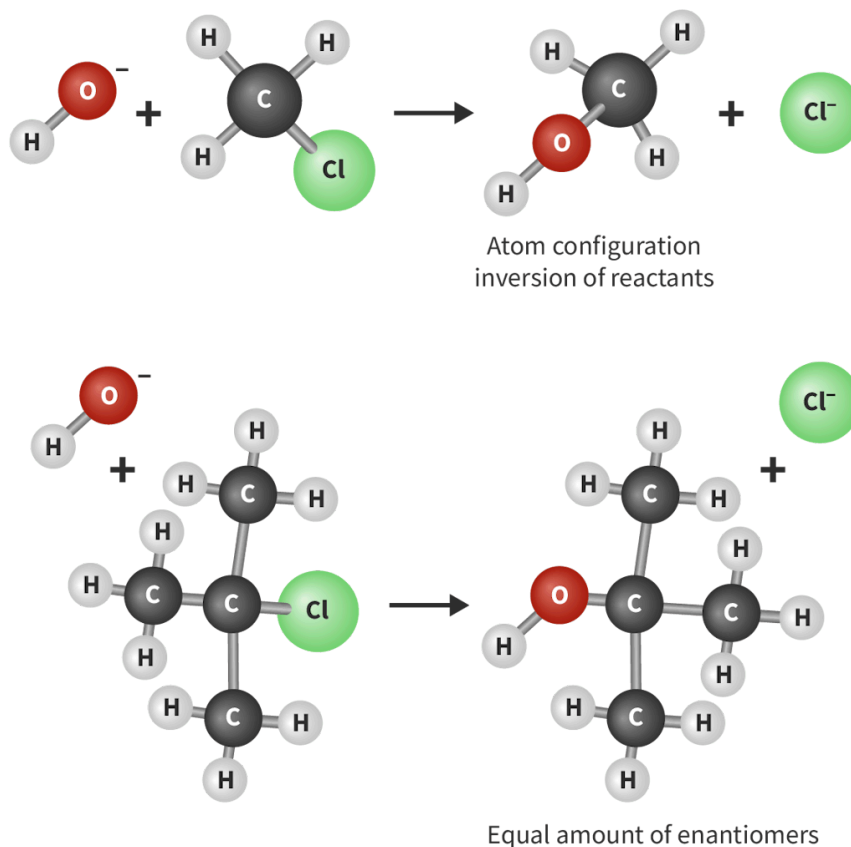


Figure 1. Alcohol formation from different types of halogenoalkanes through different mechanisms.

 More information for figure 1

The image shows two nucleophilic substitution reactions of halogenoalkanes with hydroxide ions, illustrating different mechanisms and outcomes.

1. Top Reaction:
2. Reactants: A chloromethane molecule (CH_3Cl) and a hydroxide ion (OH^-).
3. Process: Hydroxide ion attacks the carbon atom bonded to chlorine in chloromethane, resulting in inversion of atom configuration.
4. Products: Methanol (CH_3OH) and a chloride ion (Cl^-).
5. Note: The reaction results in atom configuration inversion of the reactants.
6. Bottom Reaction:
7. Reactants: A 2-chloro-2-methylpropane molecule ($(\text{CH}_3)_3\text{CCl}$) and a hydroxide ion (OH^-).
8. Process: Hydroxide ion attacks the carbon atom bonded to chlorine, resulting in a different intermediate.
9. Products: Tertiary butyl alcohol ($(\text{CH}_3)_3\text{COH}$) and a chloride ion (Cl^-).
10. Note: The reaction leads to equal amounts of enantiomers due to the formation of a racemic mixture.

[Generated by AI]

The nature of nucleophilic substitution reaction mechanisms depends on the types of halogenoalkane involved in the reaction. [Section R3.4.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/print/)
([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/print/)

[753301/book/nucleophilic-substitution-reaction-id-43490/](#)) introduces the way in which the electron pairs are moved from the nucleophile attacking the partially positively charged carbon atom, causing the halogen atom to leave the group as a leaving group. As a result, the carbon–nucleophile bond and halogen ion are formed. However, the process of this reaction occurs in a series of elementary steps, and two major steps are identified: breaking the C–X bond (departing of the halogen) and forming the C–Nu bond (nucleophilic attack).

The step in which the C–X bond breaks between the carbon atom and the halogen atom results in the departure of the halogen atom, often referred to as the ‘leaving group’. The other step is when the nucleophile attacks the partially positive carbon atom as it has a polar bond with the halogen atom. The nucleophile is an electron rich species, attacking the carbon atom by donating a pair of electrons to form a new bond with the carbon atom. Which step happens first? What factors might determine the order of the steps?

The structure of the carbon atom to which the halogen group is bonded (primary, secondary or tertiary) (see [section S3.2.6](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/>)) plays a significant role in identifying whether the reaction will proceed through different mechanisms, S_N1 or S_N2 reaction mechanism. S_N means substitution nucleophilic: the number 1 means unimolecular and number 2 means bimolecular. How does the structure of the halogenoalkanes affect the reaction mechanism? The two reaction mechanisms differ in their steps in which the leaving group departs. The nature of the reaction mechanisms depends on several factors such as the type of halogenoalkane involved and the structure of the carbon–halogen bond.

S_N2 (nucleophilic bimolecular substitution)

This reaction mechanism occurs by the breaking of the carbon–halogen bond due to nucleophilic attack and the simultaneous departure of the leaving group. This means that the nucleophilic substitution proceeds in one step and occurs in primary halogenoalkanes. Primary halogenoalkanes have a structure in which the carbon–halogen bond is attached to only one other carbon atom. This results in less steric hindrance as hydrogen atoms are relatively small. The space around the carbon–halogen bond is unobstructed as it has only one carbon atom attached, making it easily accessible to the nucleophile.

Study skills

The presence of three alkyl groups in the tertiary halogenoalkanes means a nucleophile may bump into an electrophile rather than react, and is called steric hindrance. Therefore, the carbon–halogen bond is broken in the first step forming the carbocation intermediate to proceed with the reaction. Below is an interactive artwork that compares the steric hindrance experienced by primary, secondary and tertiary halogenoalkanes when undergoing nucleophilic attacks. Click on the arrows in **Interactive 1** to view the different stages.

Interactive 1. Steric Hindrance Hinders the Nucleophilic Attacks in Primary, Secondary and Tertiary Halogenoalkanes.

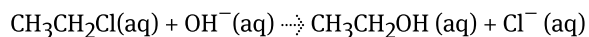
 More information for interactive 1

The interactive illustrates the effect of steric hindrance on nucleophilic substitution in primary, secondary, and tertiary halogenoalkanes in four sections. The common text on the left of each section reads as follows. Steric hindrance hinders the nucleophilic attacks in primary, secondary, and tertiary halogenoalkanes. Section 1. At the top, molecular structures of the halogenoalkanes depict the increasing steric hindrance as the number of alkyl groups around the carbon-halogen bond increases. The bottom section uses space-filling models to emphasize how alkyl groups obstruct nucleophilic attack. In primary halogenoalkanes, nucleophiles can attack with relative ease, whereas secondary halogenoalkanes experience some restriction due to bulkier alkyl groups. In tertiary halogenoalkanes, steric hindrance is most pronounced, effectively preventing nucleophilic attack.

Section 2. Primary halogenoalkane. The diagram focuses on the steric hindrance in primary halogenoalkanes during nucleophilic substitution. The left side presents a ball-and-stick molecular structure, showing the halogenoalkane with a nucleophile (Nu) approaching the carbon bonded to a chlorine atom. Steric hindrance from surrounding hydrogen atoms is indicated. On the right, a space-filling model visually emphasizes how the alkyl group partially obstructs the nucleophilic attack. Despite some hindrances, primary halogenoalkanes are still relatively accessible to nucleophiles compared to their secondary and tertiary counterparts. Section 3. This diagram illustrates the steric hindrance in secondary halogenoalkanes during nucleophilic substitution. On the left, a ball-and-stick molecular structure depicts a secondary halogenoalkane with a nucleophile (Nu) approaching the carbon bonded to a chlorine atom. The presence of two alkyl groups increases steric hindrance, restricting nucleophilic attack. On the right, a space-filling model visually highlights how the bulky alkyl groups further obstruct the nucleophile's access to the reactive carbon, making substitution more challenging compared to primary halogenoalkanes.

Section 4. This diagram illustrates steric hindrance in tertiary halogenoalkanes during nucleophilic substitution. On the left, a ball-and-stick molecular structure represents a tertiary halogenoalkane, where a nucleophile (Nu) attempts to approach the central carbon bonded to a chlorine atom. The three surrounding alkyl groups create significant steric hindrance, preventing nucleophilic attack. On the right, a space-filling model further emphasizes how the bulky alkyl groups completely block access to the reactive carbon, making direct nucleophilic substitution highly unlikely.

For example, in the reaction between chloroethane and aqueous sodium hydroxide:



the mechanism of this reaction is a one-step concerted reaction in which the hydroxide ion as the nucleophile is attracted to a slightly positive carbon and forms an unstable transition state (**Interactive 1**). The electronegativity of the leaving group plays a significant role in this process. As the leaving group is more electronegative than carbon, it has a greater ability to attract a pair of electrons towards itself, causing the leaving group to depart leaving the carbon atom to be electron-deficient and thus more positively charged. Also, a one-step concerted reaction refers to a type of reaction mechanism, which means a chemical reaction in which all bond breaking and bond making occur in a single step without any intermediate stages or unstable products. All the atoms involved in the reaction are rearranged and their positions generate the products at the same time.

The chemical structure illustrated in the square bracket in **Interactive 2** is not considered an intermediate product. It continues with the departure of chloride atoms as a chloride ion and produces alcohol. Click on the buttons to view the mechanism stages.

Interactive 2. The S_N2 Reaction Mechanism.

 More information for interactive 2

The interactive displays The S_N2 reaction mechanism through seven slides. Slide 1: A carbon atom is bonded to two hydrogen atoms in wedge and dash bonds. The carbon atom is also bonded to one methyl group (CH₃) and a chlorine (Cl) atom. The carbon atom has a positive charge and it is mentioned using δ⁺ and the chlorine atom has a negative charge and is represented as δ⁻. The bond between central carbon and chlorine is pointed using an arrow as a polar bond. The electrons in the polar bond are mentioned as two dots, one from carbon and one from chlorine.

Slide 2: The same structure is redrawn. An arrow is drawn to the carbon atom with positive charge, indicating that it is an Electron-deficient carbon atom.

Slide 3: A hydroxy group (OH) with a lone pair of electrons attacks the electron-deficient carbon atom from the left side.

Slide 4: The Transition state of the reaction is drawn. The hydroxy group is now bonded to the central carbon along with two hydrogen atoms, a methyl group and chlorine. The bonds of the hydroxy group and chlorine atom to the central carbon are in dashed lines. The whole structure is drawn inside a bracket with a negative charge on top of the bracket.

Slide 5: The chlorine atom leaves from the transition state. This is indicated by an arrow drawn from the bond towards the chlorine atom.

Slide 6: The hydroxy group, OH, is now bonded to the central carbon atom, and the chlorine ion is given as the product with four pairs or lone pairs and a negative charge.

Slide 7: The overall chemical reaction. The first structure on the left shows a hydroxide ion, OH⁻, attacking a carbon bonded to a chlorine atom, the carbon atom has δ^+ charge, and the chlorine atom has δ^- charge. A curved arrow indicates bond breaking between the carbon and chlorine. The middle structure represents the transition state, where the hydroxide and chlorine are both partially bonded to the carbon. The right structure shows the final products, where the hydroxide is fully bonded to the carbon, forming an alcohol, and the chloride ion, Cl⁻, is released. The first step is labelled slow, and the second step is labelled fast.

The S_N2 reaction mechanism is often called a ‘backside attack’ as the reaction occurs because the nucleophile attacks the partially positively charged carbon atom that is located on the opposite side of the halogen atom. The nucleophile has a lone pair of electrons, making it electron rich and the leaving group is typically an atom or group with a high electronegativity. What kind of interactions would you expect between them as they are both electron-rich species and negatively charged?

They both will experience electrostatic repulsion between them. This causes the nucleophile to be repelled by the electron-rich leaving group if it approaches from the same side. The nucleophile approaches the positively charged carbon atom from the side opposite to the leaving group to minimise repulsion.

Once the nucleophile attacks the partial positively charged carbon atom, it displaces the leaving group and simultaneously forms a new bond. This process can be visualised as a ‘bumping’ effect similar to Newton’s cradle, as the nucleophile pushes the leaving group out of the molecule. The incoming nucleophile represents the striking ball and the leaving group acts as the ball on the end. When the nucleophile approaches the partial positively charged carbon atom, the momentum is gained and transferred to the leaving group, causing it to be ‘bumped’ out of the molecule.

Furthermore, the backside attack by the nucleophile causes the inversion of the configuration in which the other atoms around carbon are being arranged, because the nucleophile needs to pass through the centre of the existing groups surrounding the carbon where both the nucleophile and the leaving group are partially bonded to the carbon. At this stage, the carbon atom configuration is not yet lost as the bond is not fully formed or broken. The new bond is formed when the leaving group completely leaves. The S_N2 reaction mechanism is said to be stereospecific due to the three-dimensional arrangement of reactants being converted into the three-dimensional stereoisomeric products. The inversion of the configuration can be visualised as an umbrella that is being blown inside-out (**Figure 2**).

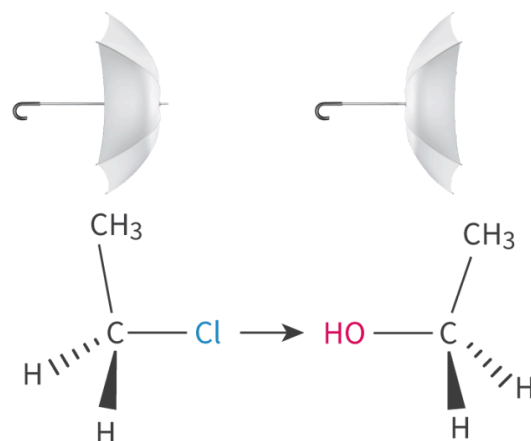


Figure 2. The inversion of atoms configuration around the atom carbon during the S_N2 reaction.

 More information for figure 2

The image is a diagram illustrating the S_N2 reaction mechanism. It features two configurations of a methyl chloride (CH_3Cl) molecule transforming into a methyl alcohol (CH_3OH) molecule, symbolizing the inversion process around the central carbon atom. The drawing includes an umbrella on each side to metaphorically represent this inversion, which appears as if an umbrella is being turned inside-out. On the left, a chlorine atom (Cl), indicated by blue text, is bonded to the carbon, with three hydrogen atoms (H) and one methyl group (CH_3) connected to it in a tetrahedral arrangement. An arrow points to the right, where a hydroxide group (OH), indicated by red text, takes the place of the chlorine atom, maintaining the tetrahedral geometry but inverted. This represents the nucleophilic substitution that characterizes the S_N2 reaction.

[Generated by AI]

The S_N2 reaction mechanism is a single-step reaction in which the rate of reaction is significantly influenced by the concentration of both the halogenoalkane and the nucleophile. It means that if the concentration of either reactant is doubled, the rate of reaction will also double. This is the important characteristic of the S_N2 reaction mechanism, where both reactants are essential for the reaction to occur. The S_N2 reaction is a bimolecular reaction where the rate of reaction is dependent on the collision between two molecules (nucleophile and halogenoalkane). Thus, it can be described as a first-order reaction with respect to both the halogenoalkane and the nucleophile (second-order overall) (section R2.2.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/>)). The reaction rate between halogenoalkane and nucleophile is expressed as:

$$\text{Rate} = k[\text{halogenoalkane}][\text{nucleophile}]$$

The energy profile for the reaction between chloroethane and the hydroxide ion is illustrated in **Figure 3**. A weak transition state is formed in which a weakened carbon–chlorine bond is broken simultaneously with the formation of a new bond between carbon and oxygen atoms that requires a high activation energy.

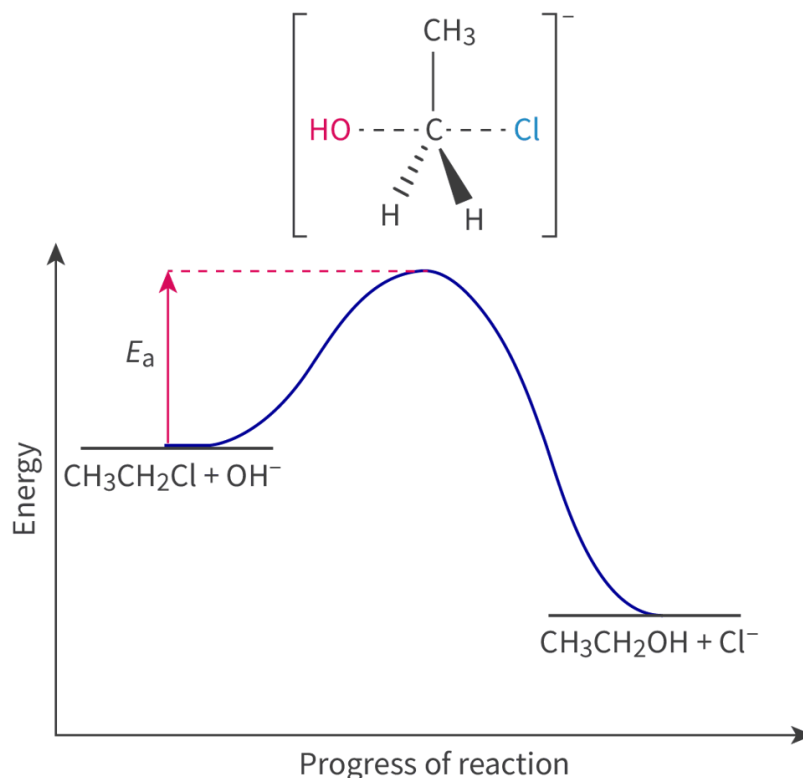


Figure 3. S_N2 reaction energy profile.

 More information for figure 3

The image is a graph representing the energy profile of an S_N2 reaction involving chloroethane (CH₃CH₂Cl) and hydroxide ion (OH⁻). The X-axis is labeled 'Progress of reaction,' while the Y-axis is labeled 'Energy.' The graph depicts a curve with an initially low energy level where CH₃CH₂Cl and OH⁻ reactants are present. This is followed by a peak representing the transition state, where the activation energy (E_a) is measured. At this point, a temporary complex is formed with partial bonds involving carbon (C), hydroxide (OH), and chlorine (Cl) atoms. The high energy indicates the energy required to break the carbon-chlorine bond while forming a new bond between carbon and oxygen. Once the transition state is surpassed, the energy decreases to a lower level, where the products CH₃CH₂OH (ethanol) and Cl⁻ are formed. This illustrates the stability of the products compared to the initial reactants.

[Generated by AI]

S_N1 (nucleophilic unimolecular substitution)

This is different from the previous nucleophilic substitution reaction mechanism, as the S_N1 reaction involves two steps and occurs in tertiary halogenoalkanes. Why is the S_N1 reaction more likely to occur? Would there be any room for the nucleophile to attack in the case of tertiary halogenoalkanes?

In the case of tertiary halogenoalkanes, the nucleophile can only attack the halogenoalkanes if the leaving group has left the molecule. It is started by dissociating the leaving group from the molecule to create room and a position for the nucleophile to occupy.

The first step that occurs is the formation of a carbocation: a carbon with a positive charge through the breaking of its carbon–halogen bond in which an electron pair between these two atoms is taken by the halogen group and then forms a carbocation intermediate and halide ion. For example, consider a reaction between 2-chloro-2-methylpropane and aqueous sodium hydroxide. The first step of this reaction results in the formation of a carbocation intermediate with a temporary positive charge and chloride ion as illustrated in **Interactive 3**. Click on the button to view the different stages.

Interactive 3. The First Step of the S_N1 Reaction Mechanism.

 More information for interactive 3

An interactive displays the first step of the S_N1 reaction mechanism in three steps in three slides.

Slide 1: It illustrates the tertiary carbon atom is bonded to three methyl groups (CH₃) and a chlorine atom (Cl). The carbon is marked as partially positive (δ^+) and the chlorine is partially negative (δ^-). A curved arrow points from the bond between carbon and chlorine to the chlorine atom, indicating the departure of the chloride ion (Cl⁻), forming a carbocation intermediate.

Slide 2: In this step, a tertiary carbocation is formed where a central carbon, bonded to three methyl groups (CH₃), carries a positive charge (C⁺). A chloride ion (Cl⁻) with a full octet of electrons is shown as the leaving group.

Slide 3: In this step, a tertiary alkyl chloride undergoes ionization: the central carbon, bonded to three methyl groups and a chlorine atom, shows partial charges (C δ^+ and Cl δ^-). A curved arrow points from the C–Cl bond to the chlorine, indicating heterolytic cleavage. The reaction is labeled as 'slow,' forming a carbocation intermediate (C⁺) and a chloride ion (Cl⁻).

The second step of this reaction mechanism involves a nucleophilic attack on a carbocation intermediate resulting in a new bond formation. The hydroxide ion donates a pair of electrons to the electron-deficient carbon atom, leading to the formation of 2-methyl-propan-2-ol, which is a tertiary alcohol (see [section S3.2.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/>)) as shown in **Interactive 4** next. Click on the button to view the stages.

Interactive 4. The Second Step of the S_N1 Reaction Mechanism.

 More information for interactive 4

An interactive displays the second step of the S_N1 reaction mechanism in three steps in three slides.

Slide 1: In this interactive, a hydroxide ion (OH⁻) donates a lone pair to a tertiary carbocation, forming a new C–O bond. The diagram shows the curved arrow representing a nucleophilic attack on the positively charged carbon center.

Slide 2: Product formed in the second step of the S_N1 reaction mechanism. A tertiary carbon atom is bonded to three methyl groups and one hydroxyl group (OH), indicating the nucleophile (OH⁻) has successfully bonded to the carbocation to form the final alcohol product.

Slide 3: The interactive showing a hydroxide ion (OH⁻) attacking a tertiary carbocation through nucleophilic attack. A curved arrow points from the lone pair on the oxygen to the positively charged carbon. The fast reaction leads to the formation of a tertiary alcohol product.

The three alkyl groups attached to the carbon atom stabilise the carbocation in the tertiary halogenoalkanes. In the case of carbon–carbon bonds, the similar electronegativities cause the bonds to be non-polar. As a result, the positive charge of the carbocation due to the electron pair removal can be shared across all three carbon–carbon bonds in the molecule. Therefore, tertiary halogenoalkanes are more stable compared to secondary halogenoalkanes which are more stable than primary halogenoalkanes. This stabilising effect perseveres for long enough time enabling the carbocation to proceed to the second step of the reaction.

Nature of Science

Aspect: Theory

Mechanistic models such as S_N1 and S_N2 are used to describe chemical reactions that involve nucleophilic substitution. Their mechanisms are widely used in the pharmaceutical and food industry. The choice between

the S_N1 and S_N2 mechanisms depends on the aim of proceeding with the reaction. The model of these reaction mechanisms helps predict the characteristics of species involved and other factors that influence the outcome of the reaction, such as the nature of species involved, the solvent and so on, on the rate of nucleophilic substitution. The theories about these two mechanisms only can be predicted and developed by collecting data including information about both reactant and product stereochemistry.

The concentration of which reactant determines the rate of the S_N1 reaction mechanism? Think about your answer before revealing the solution.

The rate of the S_N1 reaction mechanism is determined by the concentration of the halogenoalkane because the nucleophile does not participate in the rate-determining (slowest) step (see [section R2.2.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/)).

The reaction rate is expressed here.

$$\text{Rate} = k[\text{halogenoalkane}]$$

The energy profile for this reaction is illustrated in **Figure 4**. The activation energy in the first step is relatively high compared to the activation energy in the second step. This occurs due to the initial stage involving an endothermic process in which the bonds are broken, while in the second step there is a strong attraction between the oppositely charged ions (carbocation and halide ion).

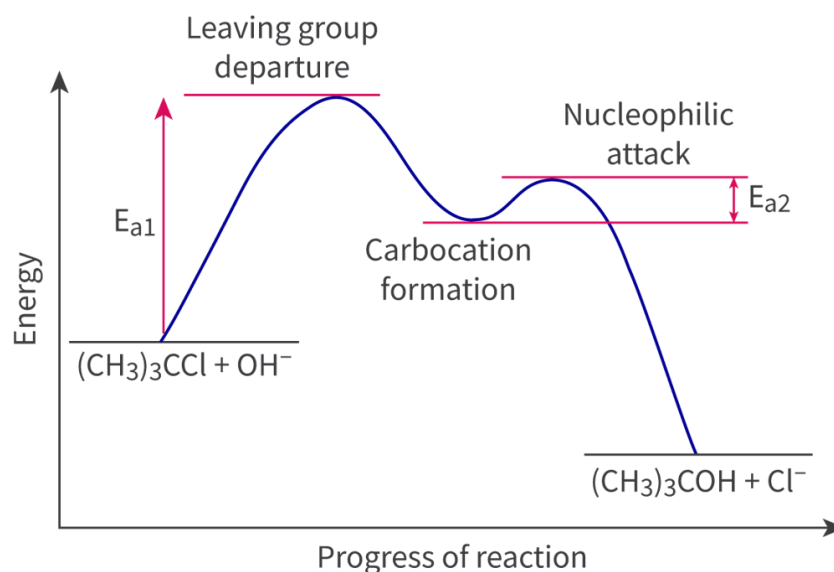


Figure 4. The second step of the S_N1 reaction mechanism.

 More information for figure 4

This is a graph representing the energy profile of an S_N1 reaction mechanism. The X-axis is labeled as "Progress of reaction" and the Y-axis is labeled as "Energy." The graph starts with the reactants $(CH_3)_3CCl + OH^-$. As the reaction progresses, there is an initial rise in energy marked as the leaving group departure, labeled with an activation energy E_{a1} . This is an endothermic process where bonds are broken. Following this, there is a decrease in energy which corresponds to the carbocation formation. The

energy reaches a peak again, indicating the nucleophilic attack, which has a smaller activation energy E_{a2} compared to E_{a1} . Finally, the product of the reaction is $(CH_3)_3COH + Cl^-$.

[Generated by AI]

Because the carbocation intermediate that is formed in the S_N1 reaction does not show stereospecificity, it allows the nucleophile to approach from any side with equal probability. This results in the formation of a mixture of optical isomers known as a racemic mixture. In [section S3.2.7b](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-optical-isomerism-hl-id-45667/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-optical-isomerism-hl-id-45667/>), a racemic mixture described as being optically inactive and having a 1:1 enantiomer mixture.

Study skills

The reaction mechanism of secondary halogenoalkanes cannot be explained in the two reaction mechanisms. This type of halogenoalkane commonly undergoes a mixture of both mechanisms (S_N1 and S_N2). The preference for a secondary halogenoalkane mechanism such as S_N2 or S_N1 depends on several factors such as the nature of nucleophile and steric hindrance. For example, if the nature of nucleophile is weak or non-basic, the S_N1 is favoured as the nucleophile is not efficient enough to displace the leaving group directly. A strong nucleophile is required as it will more readily donate its electron pair and overcome the steric hindrance to replace the leaving group in a single step. Also, the steric hindrance of the secondary halogenoalkanes can hinder the direct nucleophile attack making the S_N1 more favourable.



Exercise

Click a question to answer



1. 2-bromo-2-methylpropane will form a tertiary _____ when it reacts with aqueous sodium hydroxide as it is a more stable intermediate. Alcohol is then formed as a result of further attack by the nucleophile on the intermediate.



Check answers

Higher level (HL)

The identity of halogens as a leaving group

The reaction rate of both S_N1 and S_N2 reactions can increase if it involves a good leaving group. The rate of reaction will be relatively high if the rate-determining step occurs quickly and this is determined by the bond breaking of the carbon–halogen bond heterolytically. The weaker the carbon–halogen bond, the quicker bond breaking will be in the higher rate of reaction.

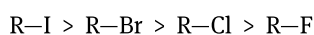
The fluorine atom that is attached to the carbon of alkane is difficult to react with because the bond between these two atoms is very strong. As you move down the group, the size of halogen atoms increases resulting in a weaker and longer carbon–halogen bond. Therefore, the halogen atom that is easier to dissipate the negative charge is iodine as it is bigger and causes weaker carbon–halogen bonds. Additionally, the polarity of the bond in iodoalkanes is the least polar due to both carbon and iodine having the same electronegativity values.

In **Table 1** is the carbon–halogen bond enthalpy data provided in [section 12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet.

Table 1. Enthalpy data for carbon halogen bonds.

Bond	Bond energy (kJ mol^{-1})
C—F	492
C—Cl	324
C—Br	285
C—I	228

The order at which a carbon–halogen bond ceases to break (most reactive to least reactive) is:



In **Interactive 5**, drag and drop the rate determining step, mechanism, order of reaction, reaction rate, carbocation stability, effects of nucleophile and stereochemistry into the correct locations in the table.

	SN1	SN2
Rate determining step		
Mechanism		
Order of reaction		
Reaction rate		
Carbocation stability		
Effects of nucleophile		
Stereochemistry		

Second-order	Rate = $k[\text{halogenoalkane}][\text{nucleophile}]$	$\text{R-X} + \text{Nu}^- \rightarrow \text{R-Nu} + \text{X}^-$
Rate = $k[\text{halogenoalkane}]$	Tertiary	One step
Inversion	Two step	No effect
First-order	Faster rate with stronger nucleophiles	Racemic mixture
$\text{R-X} \rightarrow \text{R}^+ \rightarrow \text{R-Nu}$		Primary

✓ Check

Interactive 5. A comparison of $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions.

Theory of Knowledge

Reaction mechanisms are supported with experimental evidence, but cannot yet be proved directly. To what extent does intuition play a role in the proposal of the steps for a mechanism? Can there be scientific truth without direct evidence?

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Designing procedures and models
 - Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 30—45 minutes

- **Activity type:** Pair activity

Your task

We will revisit the cat substitution analogy introduced at the beginning of this section. In pairs, create one analogy for each of the S_N1 and S_N2 reaction mechanisms, discuss factors affecting mechanisms and relate them to a cat substitution scenario. You can illustrate both analogies on a large piece of paper to visually represent them.

Materials

- A large piece of paper
- Markers or coloured pens/pencils.

Some factors you may include:

- Size/age and stability of the cat drawn to represent the leaving group.
- Aggressiveness of the attacking cat drawn to represent the strength of the nucleophile.
- Availability of space for the attacking cat to represent the steric hindrance.
- Persistence of the cat in representing the reaction rate (faster or slower).
- Resistance to the disturbance representing the stability of intermediates formed.

You can include any other factors in as much detail as possible.

Provide examples of reactions based on the analogies you have created and discuss them with other student pairs. Analyse whether they follow both S_N1 and S_N2 mechanisms based on the factors discussed. Summarise the key ideas addressed in the activity, and emphasise the analogies, factors and the relationship to the reaction example.

Other analogies you may want to illustrate:

- Team substitution: the captain of a sports team steps down from the team (players) and a new player wants to join them by substituting one of the existing players.
- The bus ride: a person in a crowded bus gets off at a stop. A new passenger waiting at the bus stop gets on the bus and notices a priority seat for people who need them. The new passenger taps the person sitting there to take the seat.
- You have four textbooks in a backpack/totebag and you realise that you accidentally took your classmate's book by mistake. You prepare to switch books — you remove their book from your bag, then place your book inside OR you place your book inside then remove their book.

Electrophilic addition reaction of alkenes (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.4.11) Describe and explain the mechanisms of the reactions between symmetrical alkenes and halogens, water and hydrogen halides.
- (R3.4.12) Predict and explain the major product of a reaction between an unsymmetrical alkene and a hydrogen halide or water.

One of the applications of electrophilic addition of alkenes is producing ethanol from ethene. The process involves the reaction of ethene with concentrated sulfuric acid to produce ethyl hydrogensulfate, which then is diluted with water. Ethanol then is produced through a distillation process. The sulfuric acid here is used as a catalyst and is reproduced in the process.

The analogy of electrophilic addition reactions can be displayed by imagining two parents, C_1 and C_2 have lost sight of their children, X_1 and X_2 at a busy park (**Interactive 1**). The parents are anxiously looking for their children and hold both hands tightly with each other. Meanwhile, the children are wandering around the park and also holding both hands with each other. As soon as they spot their children, the parents rush towards them with relief. However, the parents face a dilemma as they do not want to let go of each other or lose track of their children again. The eventual solution is that C_1 lets go of one of their hands and takes the of X_1 , while X_2 quickly grabs the hand of C_2 . In this sequence, the family is finally reunited. Think of it as a reaction of an electrophile with alkenes, where the parents are carbon-carbon double bonds and the children are halogen atoms. The reunion is the result of a sequence that breaks the pi bond and forms the halogenoalkane.

In **Interactive 1**, click through the images to view the stages.

Interactive 1. Electrophilic Addition Reactions Analogy.

 More information for interactive 1

An interactive with five sections with title “Electron addition reactions analogy”.

Section 1. An illustration of a park scene with multiple people engaging in various activities. In the foreground, two individuals labeled C₁ and C₂ are holding both hands on a path. To the right, several people are jogging, and two individuals are playing football. In the middle ground, a woman is walking while looking at her phone. Further back, a person in a wheelchair is accompanied by another person standing nearby. On the right side, two individuals labeled X₁ and X₂ are walking together, while a man is jogging past them. In the background, two people are lying on a mat. The park contains multiple trees and bushes, with paths branching in different directions.

Section 2. In the given park scene, the two individuals labeled X₁ and X₂ walk towards the two individuals holding hands labeled C₁ and C₂.

Section 3. Now the two individuals labeled X₁ and X₂ walk together but do not hold hands anymore as they walk towards the two individuals holding only one hand labeled C₁ and C₂.

Section 4. The four labelled individuals are walking in a line, holding hands. The formation from left to right is as follows. X₁, C₁, C₂, X₂.

This analogy is applied in reactions of alkenes that are either symmetrical or unsymmetrical in general, which is similar to that described in [section R3.4.4–5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/>) where it focuses more on how the addition reaction will reduce the degree of unsaturated compound and form a new compound by involving an electrophile. In this section, the additional reactions of alkenes focus more on the alkene structures (symmetrical and

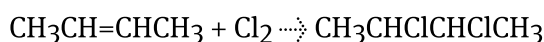
unsymmetrical), as these produce specific products. What are the differences between symmetrical and unsymmetrical alkenes? What about their products? Are they different?

Symmetrical alkenes

When alkenes undergo an electrophilic addition reaction, either end of the original alkenes are being added to by the new substituents. The products when the alkenes are symmetrical and when they are unsymmetrical are not identical. This is because each end of the double bond in symmetrical alkenes have the same substituents whereas unsymmetrical alkenes do not.

Reaction with halides

The addition of halogens, water or hydrogen halides to symmetrical alkenes produces only one type of product. For example, the reaction between but-2-ene and chlorine produces 2,3-dichlorobutane is shown in **Interactive 2**, next.



The mechanism starts when the alkene repels the electrons in the chlorine molecule, because if the double bond is an electron-rich a chlorine molecule is polarised. The chlorine molecule then is split heterolytically and forms a chloride ion causing the chlorine atom near the carbon atom to become electron deficient (an electrophile).

As a result, two π electrons move to form σ bond between carbon with the nearer chlorine atom, producing a carbocation. Therefore, chloride ions act as nucleophiles and donate a pair of electrons to the carbocation intermediate and form a coordination bond.

Click through the stages of **Interactive 2** to view the reaction mechanism.

Interactive 2. Electrophilic Addition of Chlorine to but-2-ene.

 More information for interactive 2

The interactive slideshow illustrates the electrophilic addition of chlorine to but-2-ene in six different slides. In all six slides, there is a text at the top that states “Electrophilic addition of chlorine to but-2-ene.”

Read below to learn what happens in each slide:

Slide 1:

The central structure shows but-2-ene, which consists of four carbon atoms. The double bond is located between the second and third carbon atoms. On the left of the double bond is a methyl group (CH_3 , first carbon) attached to the second carbon. On the right is another methyl group (CH_3 , fourth carbon) attached to the third carbon. Each of the two central carbon atoms in the double bond also has one hydrogen atom attached.

A chlorine molecule ($\text{Cl}-\text{Cl}$) is positioned below the double bond of the but-2-ene molecule. The chlorine atom closer to the double bond of 2-butene is labelled δ^+ (delta plus) in light blue. The second chlorine atom, farther from the double bond, is labelled δ^- (delta minus) in pink and has a pair of electrons toward it. This chlorine molecule is labelled as an “induced dipole”.

Slide 2:

This slide shows the interaction step before the reaction proceeds. The central structure shows but-2-ene and is the same as explained in slide 1. The chlorine molecule is also located below the double bond and looks similar as shown in slide 1. However, there aren't any electrons shown near the chlorine molecule in this slide. A curved arrow from the double bond points toward the δ^+ chlorine atom.

Slide 3:

Slide 3 shows the but-2-ene and Cl_2 molecule and they look similar to the ones in slide 2. The only difference is that there is a second arrow. The second arrow points from the $\text{Cl}-\text{Cl}$ bond toward the δ^- chlorine and there is a caption on the right side of Cl_2 that reads “Split heterolytically”.

Slide 4:

This image represents the intermediate stage in the electrophilic addition of chlorine (Cl_2) to but-2-ene, showing what happens after the double bond breaks and the molecule begins to rearrange. The main structure in the image is a modified form of but-2-ene, a four-carbon molecule. The double bond between the second and third carbon atoms has been broken. The second carbon is now bonded to a hydrogen atom (H), the first carbon which is a methyl group (CH_3) and a chlorine atom (Cl) shown in blue. The third carbon is bonded to a hydrogen atom (H) and fourth carbon atom which is a methyl group (CH_3). Additionally, the third carbon now carries a positive charge ($^+$), indicating that it has become a carbocation — a carbon atom that has lost an electron. A chloride ion (Cl^-) is shown nearby, in pink with a pair of dots (representing lone pair electrons), indicating that it is negatively charged and no longer bonded to any other atom. A note to the right of the image reads: “The pi bond breaks”.

Slide 5:

This slide focuses on the interaction between a carbocation and a chloride ion (Cl^-). The slide shows the modified form of but-2-ene, the same as shown in the previous slide. However, a chloride ion (Cl^-) is shown in pink with a lone pair of electrons (two pink dots), and it is positioned close to the positively charged carbon. A curved arrow points from the lone pair of electrons on Cl^- toward the positively charged carbon, indicating that the chloride ion is donating a pair of electrons to form a new bond, stabilizing the carbocation. Slide 6:

This image shows the full reaction mechanism for the electrophilic addition of chlorine (Cl_2) to but-2-ene, presented in three distinct structures (steps) from top to bottom. In the first structure (top left), the but-2-ene molecule is shown with a Cl_2 molecule beneath the double bond. There are two curved arrow marks: one between the double bond and the δ^+ chlorine atom, and the other from the $\text{Cl}-\text{Cl}$ bond toward the δ^- chlorine.

There is an arrow mark towards the right of this structure pointing towards the next step on the top right. In the second step, the structure shown is exactly similar to the one explained in slide 5. Beneath this structure, there is an arrow pointing downwards.

Below this arrow, there is a third structure indicating the product formation. This is the final molecule in which both formerly double-bonded carbon atoms now have a chlorine atom attached: One in blue (originally from Cl_2), one in pink (originally the Cl^- ion). This forms 2, 3-dichlorobutane, the final stable product.

This reaction mechanism in the slideshow involves the three main steps of electrophilic addition: Polarization and attack on Cl_2 , carbocation formation and chloride ion approach, and final bonding to form a stable saturated product.

Through this interactive viewers gain insight into heterolytic bond cleavage, carbocation formation, and nucleophilic attack, reinforcing core concepts in organic reaction mechanisms.

Reaction with hydrogen halides

The mechanism reaction between an alkene and hydrogen halide is similar to the previous mechanism. For example, the reaction between but-2-ene and hydrogen chloride produces 2-chlorobutane as shown in **Interactive 3**.



In this reaction, but-2-ene contains a double bond, which has a strong σ bond and a weak π bond, that acts as a nucleophile donating two pi electrons to an electrophile. Which atom would be attacked first by the pi electrons in but-2-ene?

Hydrogen chloride has a polar hydrogen—chloride bond in which the hydrogen atom is an electrophilic centre. Therefore, the pi electrons from the carbon double bond in but-2-ene attack the hydrogen atom of

hydrogen chloride, leading to the formation of a new bond.

The attraction of a pair of electrons in the but-2-ene by the hydrogen chloride as an electrophile causes a pair of electrons in the covalent bond between the hydrogen and chloride atom to move onto the chloride ion. The hydrogen–chloride bond splits heterolytically as the hydrogen atom can only form one bond and forms chloride ions. As the π bond no longer exists, this allows the formation of a carbocation intermediate and chloride ion. The electron pair on the negatively charged chloride ion is attracted to the positive carbon atom in the carbocation leading to the addition product.

Click through the images to view the stages of the reaction mechanism in **Interactive 3**.

Interactive 3. Electrophilic Addition of Hydrogen Chloride to but-2-ene.

 More information for interactive 3

This interactive guides users through the stepwise mechanism of the electrophilic addition of hydrogen chloride (HCl) to but-2-ene. The process is broken into six sections, which users can explore one at a time using navigation buttons given below.

Section 1: Chemical structure of but-2-ene. The double bond between the two central C atoms to. Chemical structure of HCl. The hydrogen has a delta positive charge and is bonded to the Cl group with a delta negative sign and a pair of electrons towards it. It is labeled as dipole-dipole.

Section 2. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the Cl group with a delta negative sign.

Section 3. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the Cl group with a delta negative sign. An

arrow points from the bond to the Cl group and is labeled Split heterolytically. Section 4. Chemical structure. C1 single bond C2. C1 is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C2 has a positive charge and is single-bonded to H and CH₃ groups.

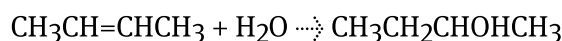
Section 5. Chemical structure. C1 single bond C2. C1 is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C2 has a positive charge and is single-bonded to H and CH₃ groups. A Cl group with a negative charge and a lone pair moves to the C2 atom.

Section 5. The overall chemical reaction. The reaction follows Markovnikov's rule, where the hydrogen (H) from HCl attaches to the carbon with more existing hydrogen atoms, forming a carbocation intermediate. The chloride group (Cl⁻) then attaches to the positively charged carbon. The final product formed is C1 single bond C2. C1 is bonded to two H atoms and a CH₃ group. C2 is bonded to a H atom, a Cl atom, and a CH₃ group.

This interactive effectively visualizes each mechanistic step, reinforcing concepts of polar bond interactions, carbocation stability, and regioselectivity in electrophilic additions.

Reaction with water

There is a similar mechanism in the reaction between alkene and water. For example, the reaction between but-2-ene and water producing butan-2-ol is shown in **Interactive 4**.



In this reaction, but-2-ene as a nucleophile attacks hydrogen atoms that have a polar hydrogen–oxygen bond. Therefore, two π electrons from the carbon double bond are used, leading to the formation of a new bond with the hydrogen atom causing the hydrogen atom bond to split heterolytically. This results in the formation of a carbocation intermediate and hydroxide ions. The hydroxide ion as a nucleophile then reacts quickly with the carbocation intermediate producing the alcohol.

Click through the stages of **Interactive 4** to view the reaction mechanism.

Interactive 4. Electrophilic Addition of Water to but-2-ene.

 More information for interactive 4

The interactive illustrates the electrophilic addition of water to but-2-ene across five sequentially numbered sections, indicated by a slide navigation bar at the bottom of the screen.

Section 1. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the OH group with a delta negative sign.

Section 2. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the OH group with a delta negative sign. An arrow points from the bond to the OH group and is labeled Split heterolytically.

Section 3. Chemical structure. C₁ single bond C₂. C₁ is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C₂ has a positive charge and is single-bonded to H and CH₃ groups.

Section 4. Chemical structure. C₁ single bond C₂. C₁ is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C₂ has a positive charge and is single-bonded to H and CH₃ groups. An OH group with a negative charge and a lone pair moves to the C₃ atom.

Section 5. The overall chemical reaction. The reaction follows Markovnikov's rule, where the hydrogen (H) from water attaches to the carbon with more existing hydrogen atoms, forming a carbocation intermediate. The hydroxyl group (OH⁻) then attaches to the positively charged carbon, resulting in the formation of butan-2-ol as the final product.

The reaction mechanism involves the initial polarization of the double bond, leading to protonation, carbocation formation, and nucleophilic attack by the hydroxide ion.

A more detailed mechanism is shown in **Figure 1**. This mechanism shows the H^+ ion acting as a catalyst, which is known as the acid-catalysed mechanism.

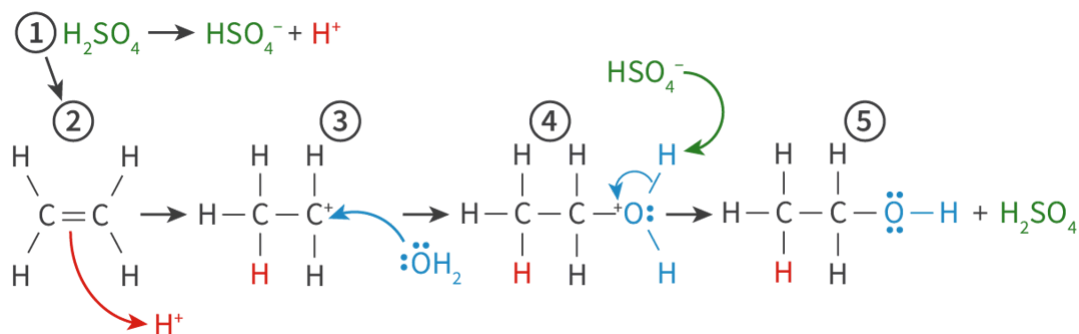
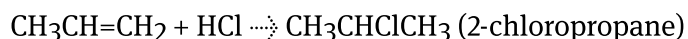


Figure 1. The acid-catalysed mechanism.

1. Sulfuric acid, H_2SO_4 , is the catalyst in the reaction and provides hydrogen ions (H^+) that act as the electrophile.
2. The H^+ ion bonds to one of the carbon atoms of the carbon-carbon double bond, forming a carbocation intermediate.
3. A water molecule bonds with the carbocation intermediate, forming a protonated ethanol molecule.
4. The HSO_4^- ion removes a proton from the protonated ethanol molecule, which reforms the sulfuric acid catalyst.
5. The product of the reaction is ethanol, $\text{C}_2\text{H}_5\text{OH}$ (in addition to the reformed sulfuric acid catalyst).

Unsymmetrical alkene reactions

On the other hand, two different products are produced when unsymmetrical alkenes react with a hydrogen halide: for example, the reaction between propene with hydrogen chloride. In this reaction, two different products are formed depending on where the halide and hydrogen atoms are attached.



The stability of carbocation determines the favoured product, which helps to predict the major product of the electrophilic addition. The more alkyl groups attached to the carbon centre, the more stable the carbocation (**Figure 2**). In [section R3.4.9–10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/>), a tertiary carbon is shown to be more stable due to more alkyl groups attached to the carbon centre.

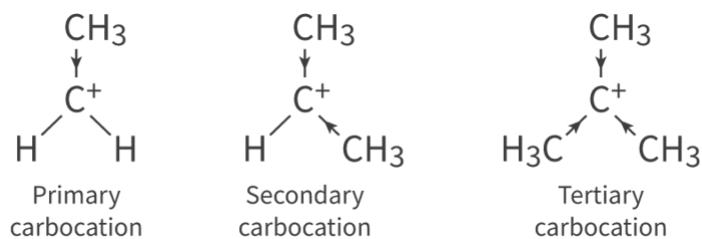


Figure 2. Stability of a carbocation.

 More information for figure 2

The image presents three structural formulas depicting a primary, secondary, and tertiary carbocation. Each structure includes a central carbon atom marked with a positive charge.

To the left, the primary carbocation is shown with the carbon atom bonded to a single methyl group (CH₃) and two hydrogen atoms (H).

In the center, the secondary carbocation features the carbon bonded to one methyl group (CH₃), one hydrogen atom (H), and another carbon bonded to a methyl group (CH₃), creating more stability than the primary.

On the right, the tertiary carbocation shows the carbon atom bonded to three separate methyl groups (CH₃), creating the most stable carbocation among the three displayed due to the highest number of alkyl groups attached.

[Generated by AI]

Study skills

Charged species are more stable when the charge can be spread out over more than one atom. A positive charge can be better dispersed over a C—C bond than a C—H bond. This is because the alkyl groups are larger and contain more electrons than an H atom, allowing their own electron density to be drawn toward the positive carbon, effectively spreading out the charge. Therefore, tertiary carbocation intermediates are more effective at spreading out the positive charge than a primary carbocation intermediate.

When propene reacts with hydrogen chloride, theoretically there will be two different products formed that are isomers of each other (**Figure 3**). This leads to two possibilities of pathways through electrophilic addition reactions.

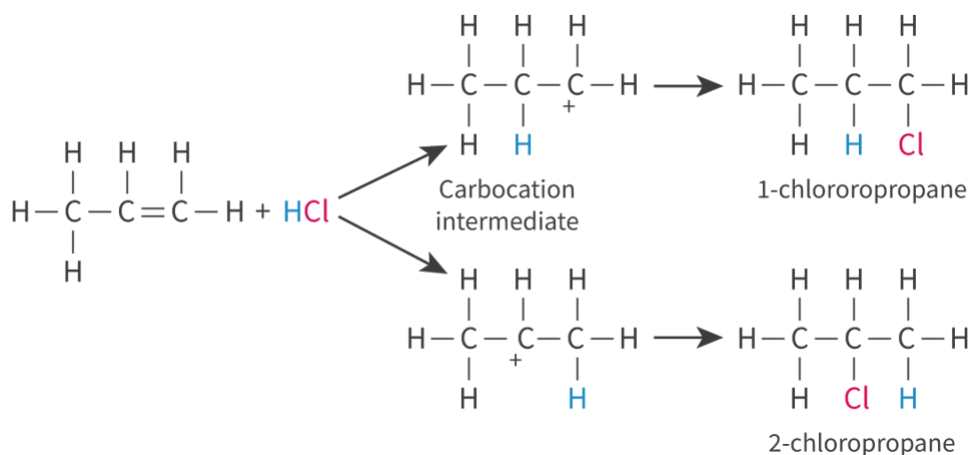


Figure 3. An electrophilic addition reaction of hydrogen chloride to propene.

 More information for figure 3

The diagram illustrates the electrophilic addition reaction of hydrogen chloride (HCl) to propene (C₃H₆). Starting at the left, the structure of propene is depicted as H₂C=CH-CH₃. Hydrogen chloride (HCl) is added to this molecule. Two possible carbocation intermediates are formed.

The first pathway splits upward, where propene reacts by adding H⁺ to the terminal carbon, creating a carbocation on the central carbon. From here, Cl⁻ attacks the positively charged center, forming 1-chloropropane (CH₃-CH₂-CH₂Cl).

The second pathway splits downward, where H⁺ adds to the central carbon, creating a carbocation on the terminal carbon. Here, Cl⁻ attacks this positively charged carbon, leading to the formation of 2-chloropropane (CH₃-CHCl-CH₃).

Both pathways are shown with arrows, indicating possible products, 1-chloropropane and 2-chloropropane.

[Generated by AI]

Which product is predominantly formed in this reaction, and why? Are both products formed in equal amounts or is one product favoured over the other? What factors may influence this and how does the presence of substituents on the propene affect the production of a major product?

The major product is 2-chloropropane as it is produced through a more stable secondary carbocation intermediate. As the 2-chloropropane is favoured in this reaction, the two isomeric products are not formed in equal amounts. The formation of major products is influenced by the nature of the carbocation intermediate. The more stable the carbocation intermediate of the reaction, the more stable the

product. The presence of substituents may alter the steric hindrance of propene. Therefore, it can create the reactivity of propene towards HCl and shift the selectivity towards the other product.

The main product of this reaction will more probably be 2-chloropropane because it is formed from a secondary carbocation that is more stable compared to the primary carbocation that is formed to produce 1-chloropropane. The accurate mechanism of the addition reaction of propene and hydrogen chloride is shown in **Figure 4**.

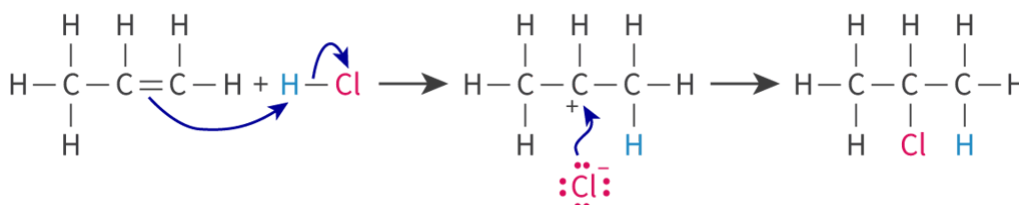


Figure 4. The mechanism of the addition reaction of hydrogen chloride to propene.

 More information for figure 4

The diagram illustrates the mechanism of the addition reaction of hydrogen chloride to propene. It shows the step-by-step formation of 2-chloropropane. The reaction starts on the left with a propene molecule (C₃H₆) depicted with its double bond between the first and second carbon atoms. The hydrogen chloride (HCl) molecule is shown next to the propene with an arrow indicating the movement of the hydrogen (H⁺) towards the double bond. This results in the formation of a carbocation on the secondary carbon as the chlorine ion (Cl⁻) is released. The chlorine ion then bonds with the carbocation forming 2-chloropropane, depicted on the right. The arrows illustrate the flow of electrons during the reaction phases.

[Generated by AI]

The product from the reaction of an unsymmetrical alkene and hydrogen halide can be predicted using Markovnikov's rule in which it is stated that a hydrogen atom will probably attach to the carbon atom that has more hydrogen atoms already attached, or 'the rich get richer'. The addition of hydrogen chloride to propene allows hydrogen to attach to carbon number 1 (**Figure 5**) as it has two hydrogen atoms attached compared to carbon number 2, which has no hydrogen atoms attached.

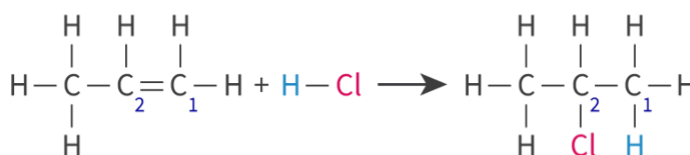


Figure 5. Predicting the major product of an unsymmetrical alkene reaction.

 More information for figure 5

The diagram illustrates a chemical reaction between propene, a type of unsymmetrical alkene, and hydrogen chloride (HCl), demonstrating Markovnikov's rule. On the left side of the equation, propene's structure shows three carbon atoms, with carbon 1 (C1) having two hydrogen atoms attached and carbon 2 (C2) having none. A hydrogen chloride molecule is added to this structure, where the hydrogen atom (H) and chlorine atom (Cl) are marked distinctly.

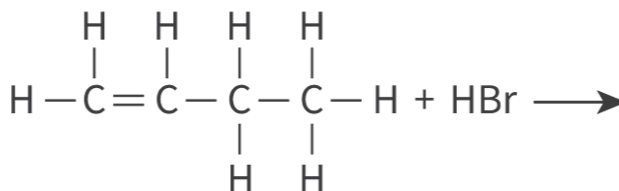
The reaction proceeds with an arrow pointing to the right, indicating the formation of the reaction products. The structure on the right shows the resulting molecule: the hydrogen atom attaches to carbon 1, which aligns with the rule that hydrogen attaches to the carbon atom with more hydrogens ('the rich get richer'). The chlorine atom attaches to carbon 2, completing the reaction product.

This diagram visually explains how Markovnikov's rule predicts the major product in the addition of HX to unsymmetrical alkenes, exemplified by the addition of HCl to propene resulting in 2-chloropropane as the product.

[Generated by AI]

Worked example 1

Predict and draw the chemical structure for the major and minor products from the reaction shown.

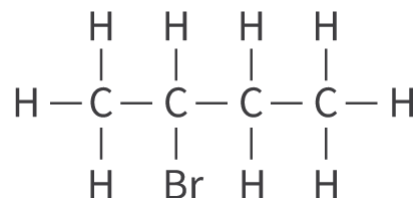


 More information

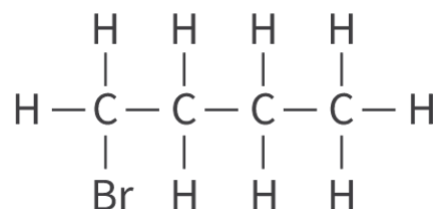
The image displays a chemical reaction with structural formulas. The left side shows the organic compound 2-methyl-2-butene, which has a double bond between two central carbon atoms, surrounded by hydrogen atoms and a methyl group. The plus sign indicates it reacts with HBr (hydrogen bromide), and an arrow points right, symbolizing a chemical reaction leading to the formation of a product.

[Generated by AI]

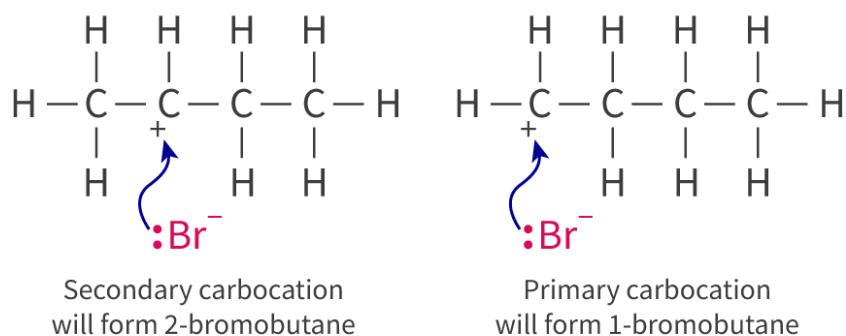
Major product



Minor product



The major product of the reaction between but-1-ene and hydrogen bromide has the Br atom attached to the second carbon in which the minor product has the Br atom attached to the first carbon atom. The secondary carbocation is formed when the hydrogen ion in HBr is attracted to the carbon atom of the double bond, and is more stable compared to the primary carbocation.





Exercise 1

Click a question to answer



1. True or false?



2-iodo-2-methylbutane is favoured in the reaction between hydrogen iodide and 2-methylbut-1-ene.



Check answers

Higher level (HL)

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:**
 - Thinking skills — Designing procedures and models
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 45 minutes + 20 minute extension
- **Activity type:** Group activity

Materials

- Modelling clay to represent atom (at least six different colours to represent the atoms of C, H, O, Br and Cl: black for carbon, white for hydrogen, orange for bromine, green for chlorine, red for oxygen, an extra colour for electrons).
- Craft wire to connect modelling clay balls to form the chemical structure and curly arrows.
- Multiple smaller balls of modelling clay to represent electrons — choose a different colour to those used for atoms. They will be attached to the atoms using the craft wire.

Your task

Construct a three-dimensional model that visually represents diverse mechanisms of electrophilic addition reactions such as halogenation, hydrogenation and hydrohalogenation for both symmetric and unsymmetric alkenes. Explain these mechanisms and the movement of electrons and curly arrows using the materials provided. By using online software or a video camera, capture a series of images step by step and create a stop motion animation/film. Then compare these reactions, and present a selection reaction mechanism and its corresponding conditions for symmetric and unsymmetric alkenes to your classmates.

Extension task

Extend your explanation by conducting further research on the application of the reactions. Connect your selected reactions to the nature of science by exploring their coverage, testing, verification and interconnections with other scientific concepts and disciplines.

Electrophilic substitution reactions of benzene (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Describe and explain the mechanism of the reaction between benzene and a charged electrophile, E^+ .
- Describe the features of benzene (C_6H_6) that make it not prone to undergo addition reactions, despite being highly unsaturated.

Section S2.2.12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/>) covers benzene, which is a compound that differs from other cyclic compounds such as 1,3,5-cyclohexatriene. Both the physical and chemical evidence of benzene are explained, and the reactivity of this compound is discussed later in this section. This might lead you to ask why is benzene not prone to undergoing such addition reactions like 1,3,5-cyclohexatriene even though it has the double bond? The benzene molecule's aromatic (ring-shaped) nature means it is very stable. Understanding the stability of benzene will help you to learn about its unique behaviour and properties, which is an important prerequisite for learning and understanding its physical and chemical evidence. How does stability play a crucial role in determining the type of reaction that benzene will undergo?

Benzene (C_6H_6) has three carbon double bonds and is highly unsaturated. However, this characteristic does not allow it to react like alkenes. Benzene usually undergoes electrophilic substitution reactions in which one or more atoms on the ring are replaced with another atom or group and will conserve the ring of benzene. On the other hand, the instability of benzene due to less common addition reactions will lead to higher energy in products than in the reactants. **Figure 1** shows the electrophilic substitution reaction between the electrophile, E^+ with benzene.

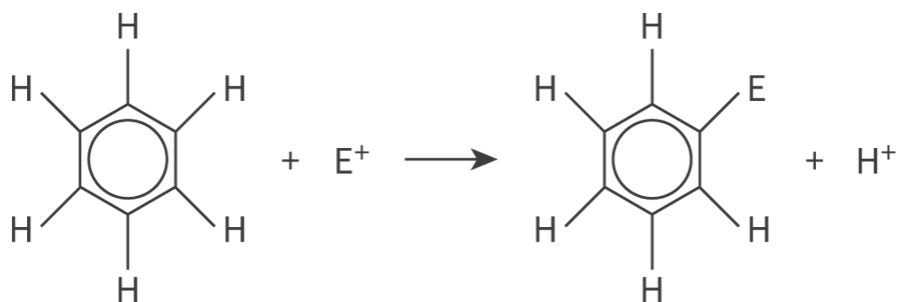


Figure 1. The electrophilic substitution reaction between the electrophile, E^+ with benzene.

 More information for figure 1

The image displays a diagram of an electrophilic substitution reaction. On the left, there is a structural formula of benzene (C_6H_6), represented as a hexagon with alternating double bonds and a circle inside, symbolizing delocalized pi electrons. Each carbon atom in the hexagon is bonded to a hydrogen atom, indicated by a line terminating with an H. To the right of the benzene, there's an arrow pointing towards another benzene structure, which now has one hydrogen replaced by an electrophile group, E. The hydrogen that was replaced is shown detached on the far right as H^+ . The reaction indicates that the electrophile (E^+) attaches to the benzene ring, replacing one of the hydrogen atoms, which becomes a separate hydrogen ion (H^+).

[Generated by AI]

Electrophilic substitution reactions of benzene involve electrophiles in which benzene plays a role as an electron-rich molecule. Benzene has a ring of six carbon atoms, each of which is bonded to other two carbon atoms and one hydrogen atom. The electrons in the bonds between the two carbon atoms are delocalised around the ring, creating a high electron density region. This leads the ring of benzene being attracted to the electrophile as an electron-deficient species seeking the electrons and forms a bond as a hydrogen departs from the ring structure. All six π electrons of benzene are available to react with an electrophile (E^+) despite their function of maintaining the stability of benzene. Benzene has a stable structure due to delocalisation energy that reduces the repulsion between the electrons. Therefore, the addition reaction is not favoured for benzene as it will interrupt the cloud of delocalised π electrons. This will lead to the instability of the benzene structure. **Figure 2** shows how benzene experiences its loss of symmetry as the positive charge is drawn inside the ring. As the hydrogen atom leaves the carbocation intermediate, the two electrons move back and reproduce the aromatic ring.

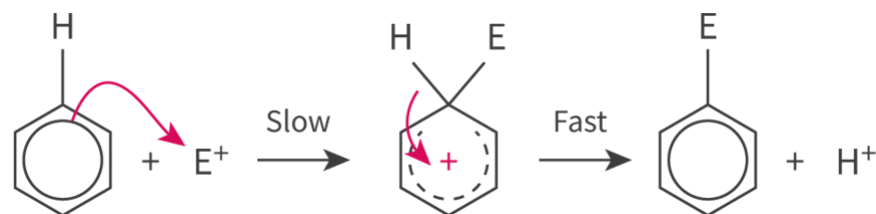


Figure 2. Electrophilic substitution of benzene.

 More information for figure 2

The diagram illustrates the electrophilic substitution reaction of benzene. It consists of three main steps:

1. The starting point is a benzene ring with a hydrogen atom (H) attached, alongside an electrophile (E^+). An arrow indicates the formation of a bond between the benzene and electrophile.
2. The next step shows an intermediate where the benzene ring has bonded with the electrophile (E), but the hydrogen is still present. Within the benzene ring, there is a positive charge signified by "+" inside a circle. An arrow shows movement of electrons inside the ring. This step is indicated as "Slow."
3. The final step shows the benzene ring restored with the electrophile attached, and the hydrogen ion (H^+) as a separate product. This is labeled as "Fast."

The diagram visually represents how the hydrogen is replaced by the electrophile, illustrating the mechanism of the reaction.

[Generated by AI]

Think of the sequence of an electrophilic substitution as a table that comfortably seats six people. You, the host (H), and your friends (1 to 5) are seated at the table, comfortable, and having a conversation over a meal. Suddenly, a new elder person, E, appears who wants to join even though there is no more space available. You share your chair with E initially but it is not comfortable, so you give up your chair for E. Now everybody at the table now feels settled and comfortable again to continue enjoying their food and conversation. E is acting as an electrophile that displaces the existence of H as a hydrogen atom. The reaction is completed when the leaving group is displaced by the electrophile.

Study skills

Enhancing specificity of mechanism drawing is crucial in addressing points that frequently negatively affect exam performance. You should note some specific aspects when drawing mechanisms:

1. Curly arrows must have a precise origin and destination: the origin being from the bonding pair or non-bonding pair that are moving and the destination being on the atom they are going to.
2. Transition states and intermediates must show correct bonding arrangements, square brackets and charge (intermediates), or dotted lines showing partially formed/broken bonds (transition states).
3. Make sure you include all final products for the overall reaction equation.

A high activation energy is required to proceed with this reaction and so the process is rather slow (**Figure 3**). The first step occurs when the electrophile is attracted to the electron cloud of the aromatic ring. The unstable carbocation is formed where both the electrophile and the leaving hydrogen are bonded temporarily to the ring. The positive charge is distributed over the ring and the incomplete circle shows that the ring system loses its symmetry. The loss of the proton generates the electrically neutral substitution product with the two electrons from the loss of the carbon–hydrogen bond moving to remake the pi ring system.

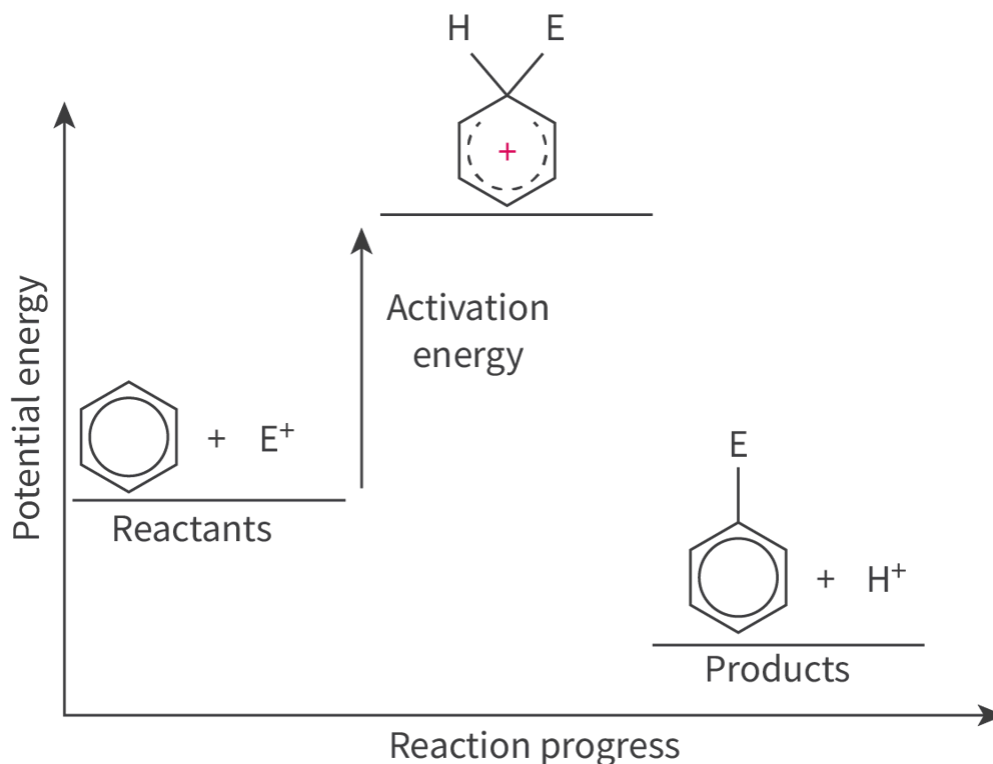


Figure 3. Energy profile of a benzene electrophilic substitution reaction.

 More information for figure 3

The image is a graph representing the energy profile of the benzene electrophilic substitution reaction. The graph has two axes: the Y-axis is labeled "Potential energy," and the X-axis is labeled "Reaction progress." The graph illustrates three major points:

1. **Reactants:** On the left side, labeled as "Reactants," there is a benzene ring with a plus symbol inside, and it is shown combining with an electrophile (E+).
2. **Activation Energy:** The middle section shows the peak of the graph, indicating the transition state. This part is labeled "Activation energy," and it features a benzene ring with double bonds represented as dashed lines. Above this structure, there's an electrophile (E) and a hydrogen atom (H) attached, symbolizing the unstable intermediate carbocation where the positive charge is distributed over the ring.
3. **Products:** On the right side, the graph levels off, indicating the final state after the reaction. Here, the benzene ring is again depicted, now firmly bonded with the electrophile (E) and a hydrogen ion (H+) is released. This section is labeled "Products."

The graph visually represents the energy changes during the reaction process, showing a significant energy barrier due to high activation energy, which corresponds to the endothermic step of forming the unstable intermediate before reaching a lower energy level at the product stage.

[Generated by AI]

Study skills

The nitration of benzene occurs through a mixture of concentrated sulfuric and nitric acids. Sulfuric acid is used to protonate the hydroxyl group in nitric acid. As a result, the nitrogen-oxygen bond in nitric acid breaks forming a water molecule and the nitronium ion is produced. This then acts as a strong electrophile that readily reacts with benzene.

Here is the chemical equation of the reaction between sulfuric and nitric acid.



Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

Your task

Together with your partner, investigate how different electrophiles can add to benzene and the resulting products. Also, compare the stability and reactivity of benzene and its derivatives.

1. Start by reviewing the structure and properties of benzene.
2. Draw the Lewis structure and the resonance structures of benzene.
3. Explain why benzene is more stable than a hypothetical 'cyclohexatriene' with three isolated double bonds.
4. Recalling the general mechanism of electrophilic substitution to benzene, draw the steps involved in the reaction of benzene with bromine in the presence of iron(III) bromide as a catalyst. Label the electrophile, the nucleophile and the intermediate complex. What is the name and structure of the product?
5. Think of other electrophiles that can react with benzene. For each of the following electrophiles, write the equation for the reaction of benzene with them and name the product. Use curly arrows to show the movement of electrons in each step.
 - Chlorine in the presence of iron(III) chloride
 - Nitric acid in the presence of sulfuric acid
 - Sulfur trioxide in the presence of sulfuric acid
 - Ethene in the presence of aluminium chloride.
6. Last, compare and contrast the stability and reactivity of benzene and its derivatives. How does the addition of an electrophile affect the aromaticity of benzene? How does the presence of an electron-withdrawing or electron-donating group on the benzene ring affect its reactivity towards electrophiles? Give examples to illustrate your points.